

AI Uniform Designer

Final Verification & Delivery Report

Independent end-to-end verification of the customer-facing AI uniform design workflow: four-concept generation, explicit concept selection, selection-scoped refinement, roster, production exports and Shopify order handoff.

Prepared for	Michael & Bryant
Prepared by	Enis Gjini
Date	August 29, 2026
Branch	main
Verified commit	1d65304 1d65304cb2981f0ab06a4591f236a5421a099f9f
Automated checks	4 of 4 command suites passing
Browser QA	32 passed / 0 failed across 7 viewports

1. Executive summary

The BESU Customs AI Uniform Designer has been taken through a full implementation and verification pass. The customer journey now works as specified: a customer describes the uniform they want, receives **four independently generated finished uniform concepts**, explicitly chooses the direction they prefer, and then refines only that chosen design before completing roster, production exports and the Shopify order handoff.

During the audit one substantive workflow defect was found and corrected. The four-concept generation capability was present in the codebase but the production generation path was routed through an older single-render function, which produced one design, selected it automatically and skipped the concept-selection step entirely. That path has been removed and every customer entry point now runs through a single canonical four-concept flow. Two further defects were corrected: production PNG/PDF exports were being rasterised to a fixed portrait canvas that distorted the landscape uniform render, and the step navigation overflowed horizontally on 360 px phones.

Release status: the workflow, responsive layout, refinement, roster, export and Shopify payload behaviour described in this report were verified directly in a real browser on the build at commit `1d65304` . The journey was verified twice: once through a deterministic preview renderer, so it could be re-run repeatedly without consuming image-model credits, and once end to end against the **live production image model**. Both runs passed all 32 checks. The live renders are shown in *Live production output*.

What was completed

- Routed all fresh customer generation through one canonical four-concept orchestration.
- Removed the retired single-collage render path and the conflicting "three labeled designs" prompt mode.
- Confirmed no concept is pre-selected, and Refine / Roster / Order stay locked until the customer chooses.
- Verified refinement and colour variation act on the selected concept only, preserving its design identity.
- Fixed production raster exports so PNG, PDF and ZIP output keeps the render's true proportions.
- Fixed horizontal overflow of the step navigation on narrow phones.
- Replaced source-pattern tests with behavioural tests that execute the real production modules.
- Added a dependency-free browser QA harness that drives the real application across seven viewports.
- Ran the full journey end to end against the live production image model, not only the preview renderer.
- Corrected two live image-model output defects: third-party brand marks appearing on the garment, and team-name spelling drift on generation and recolouring.

2. Client requirements and final status

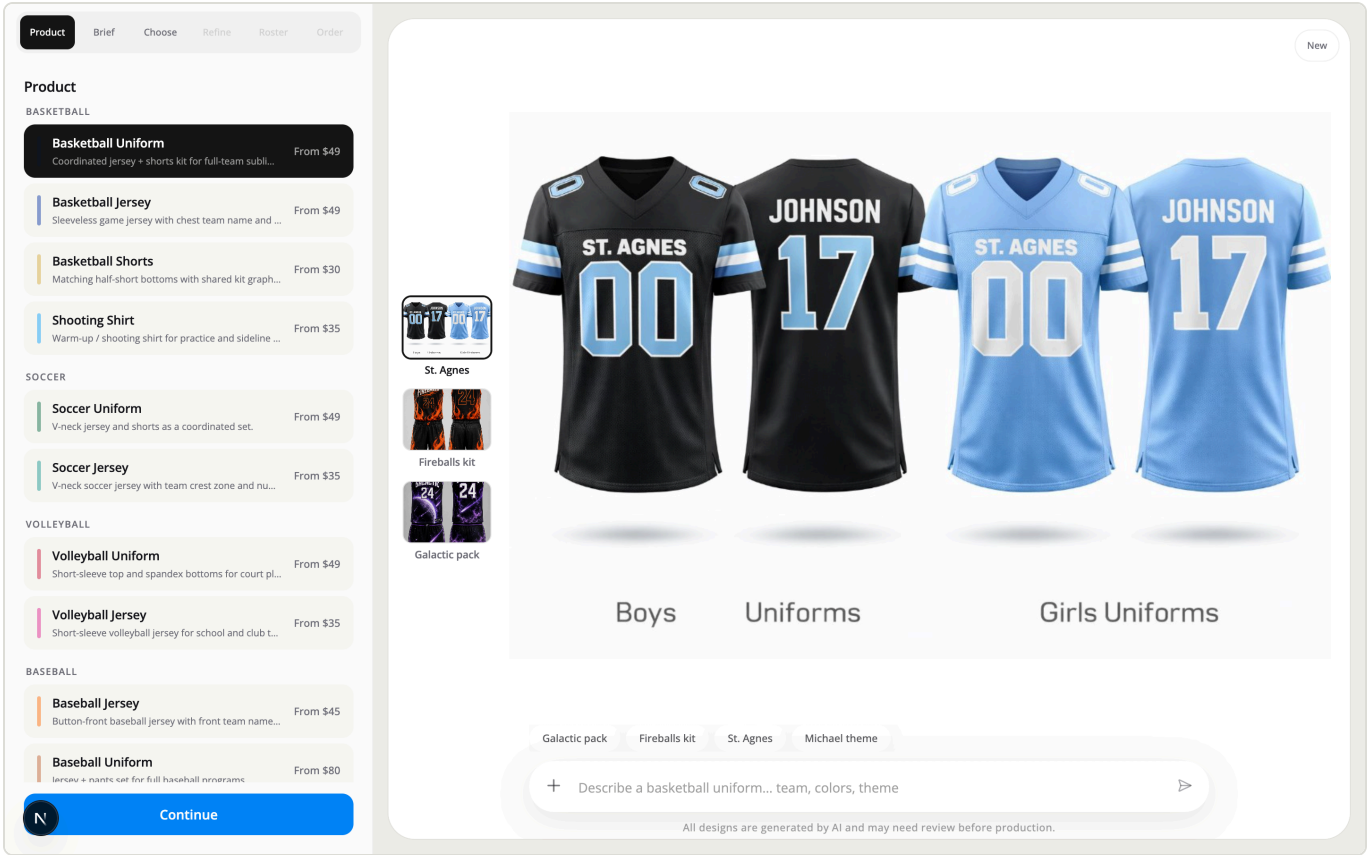
REQUIREMENT	STATUS	EVIDENCE / NOTE
Customer submits a design brief	Verified	team GALACTIC, 217 char brief

Four distinct finished uniform concepts generated	Verified	4 requests issued
All four concepts shown in one choice step	Verified	4 cards (4 visible), 4 unique renders
No concept selected automatically	Verified	Refine, Roster and Order disabled with aria-disabled=true
Refine / Roster / Order locked before selection	Verified	Tabs disabled with aria-disabled="true"
Explicit customer selection drives artwork and design ID	Verified	Cosmic Energy selected, preview + designId updated, downstream unlocked
Selection can be changed cleanly	Verified	selection switches both directions without leaking artwork or history
Natural-language refinement edits the selected design	Verified	history 1, designId preserved, 5 total generation calls
Colour variation preserves the design	Verified	palette #0A0A0A / #00A3FF / #FFFFFF, garment identity preserved
Roster add / edit / delete with independent sizes	Verified	Bryant #24 XL/XL x2, Carter #8 L/M x1
Production exports available for the current design	Verified	PNG, SVG, PDF, ZIP
Exports use the current render at correct proportions	Verified	2400x1600 raster from the current render
Shopify handoff carries the selected design	Verified	designId 1d8f6ec3-491f-4988-8b2d-624de6b1de3f, 2 items [2,3], variants 51354887192854, 51354887127318
Product catalogue intact and resets state on change	Verified	jersey <-> uniform switch clears concepts

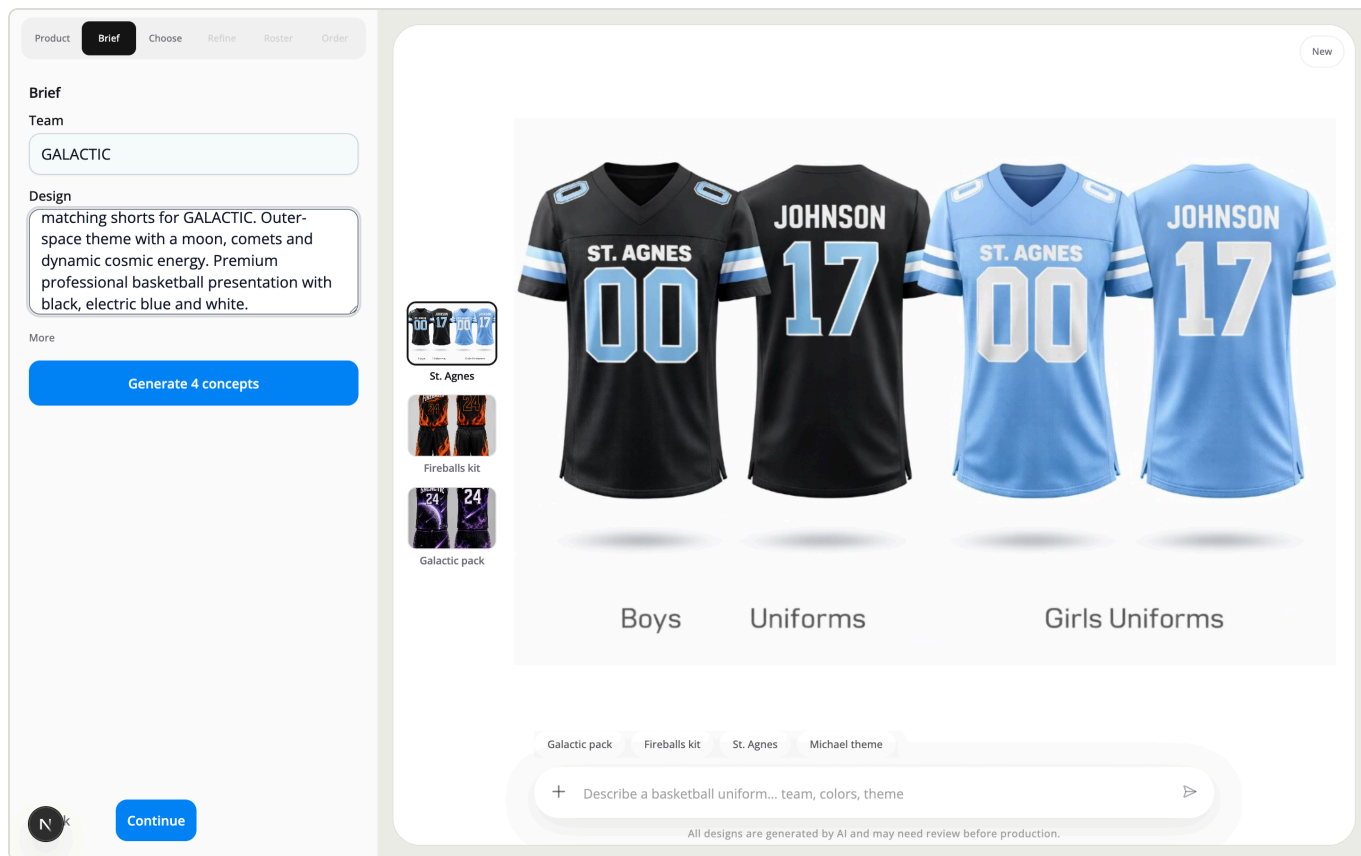
3. The four-concept workflow

The customer journey is a six-step flow: **Product** → **Brief** → **Choose** → **Refine** → **Roster** → **Order**. The Choose step is the commercial heart of the product, and it is the step that was previously being skipped.

- 1. **Product.** The customer picks the garment. Changing product clears any previous design so a stale concept can never be carried into an order.
- 2. **Brief.** The customer supplies a team name and a description of the uniform they want.
- 3. **Choose.** One brief produces four separate image-model renders, each pushed toward a different art direction. All four are presented together, none is pre-selected, and the downstream steps remain locked.
- 4. **Refine.** Once a concept is chosen, that render becomes the working design. Text refinements and colour variations are applied to it as edits, so the chosen identity is preserved.
- 5. **Roster.** Players, numbers, sizes and quantities are captured per row.
- 6. **Order.** Customer details, production exports and the Shopify handoff all read the current refined design.



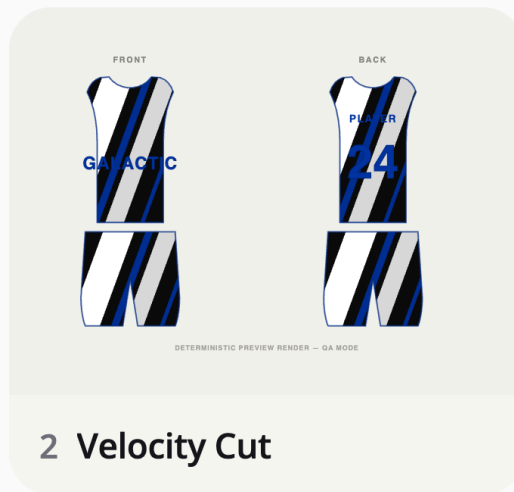
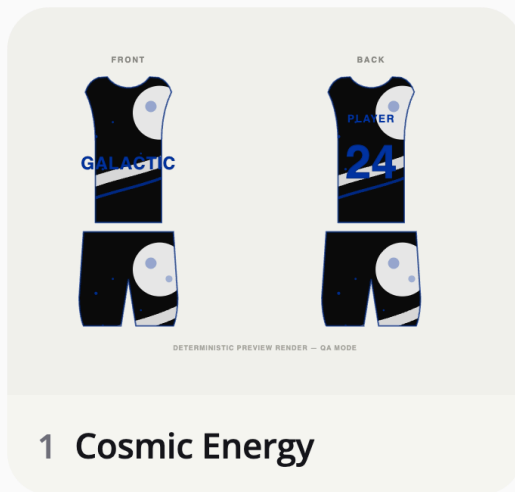
01-desktop-initial-product.png Step 1 — product selection. Eighteen production garments across seven sports; the basketball uniform is the default.



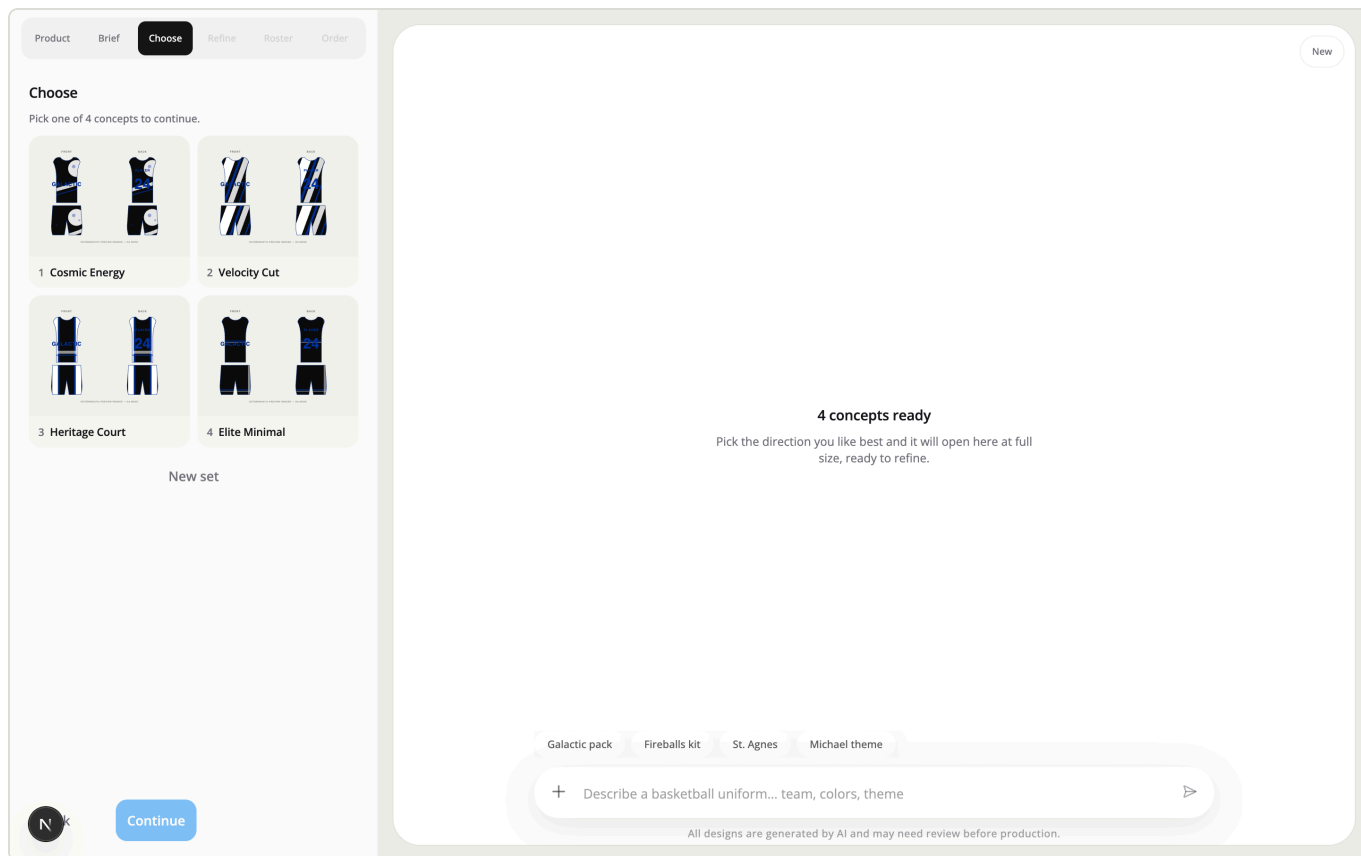
02-desktop-ai-brief.png Step 2 — design brief. Team name GALACTIC with the full cosmic uniform brief; the action is explicitly labelled "Generate 4 concepts".

4. Four concepts, presented for choice

Submitting the brief issues four independent generation requests, one per art direction. The browser QA harness counts the outbound requests, so this is measured rather than assumed: **4 requests issued**. The four resulting renders are stored as four separate concepts, each with its own design ID and asset URL.



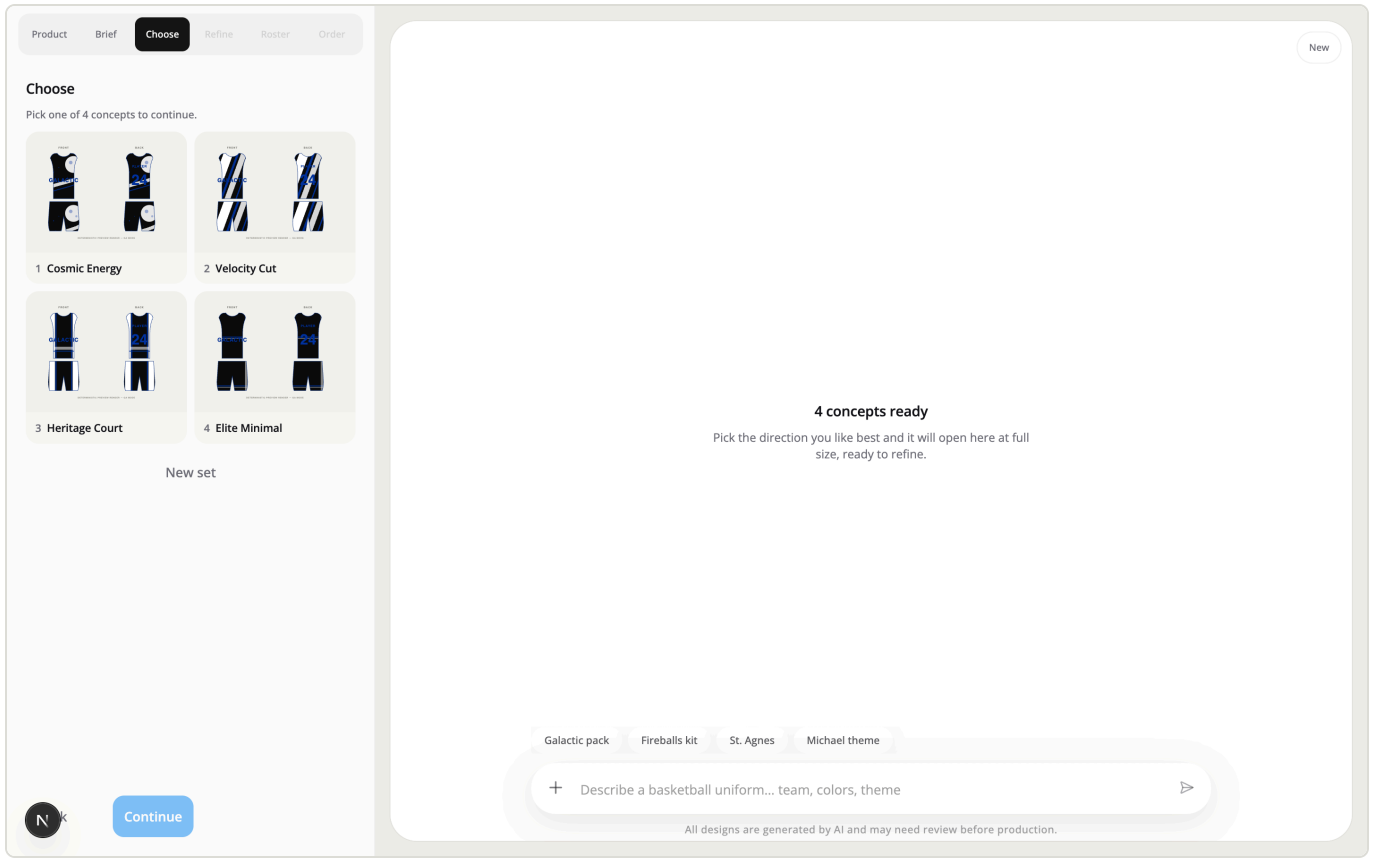
03a-desktop-four-concepts-detail.png The four concepts produced from the single GALACTIC brief, shown at full detail. Each is an independent render with its own design ID: a sleeveless jersey and matching shorts, front and back, with the team name on the chest and a clean number area on the back.



03-desktop-four-concepts.png Step 3 in context — the Choose step. All four concepts are presented together in one two-column choice grid with the instruction "Pick one of 4 concepts to continue."

Note the state of the interface in the screenshot above: nothing is selected, and the *Refine*, *Roster* and *Order* tabs are visibly dimmed. That is the required behaviour — the customer must make a choice before the flow continues.

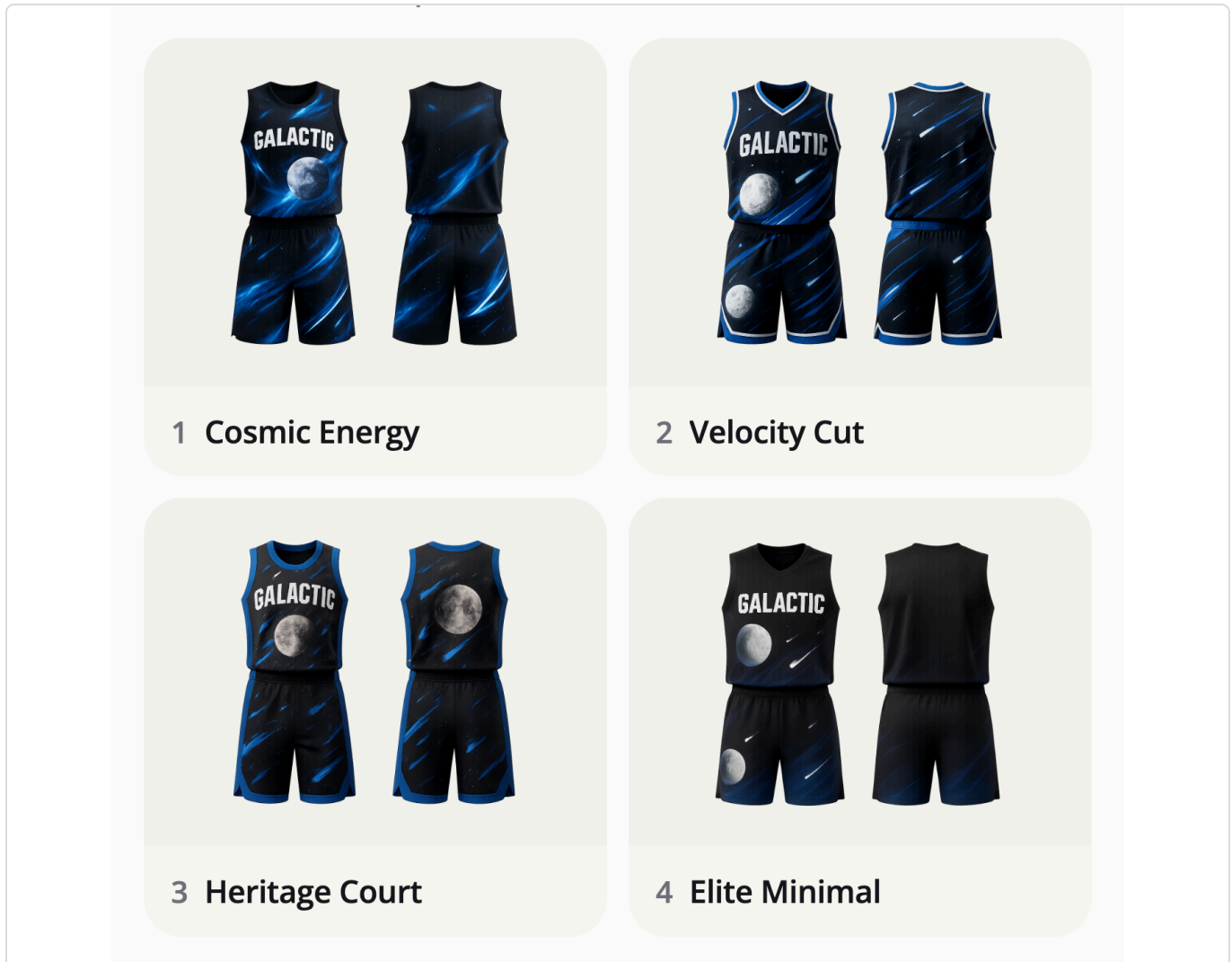
The small "deterministic preview render — QA mode" caption inside each render is part of the verification renderer used for the repeatable regression run, not the product. It is present so these screenshots can never be mistaken for production image-model output. The next section shows the same four directions rendered by the live production image model.



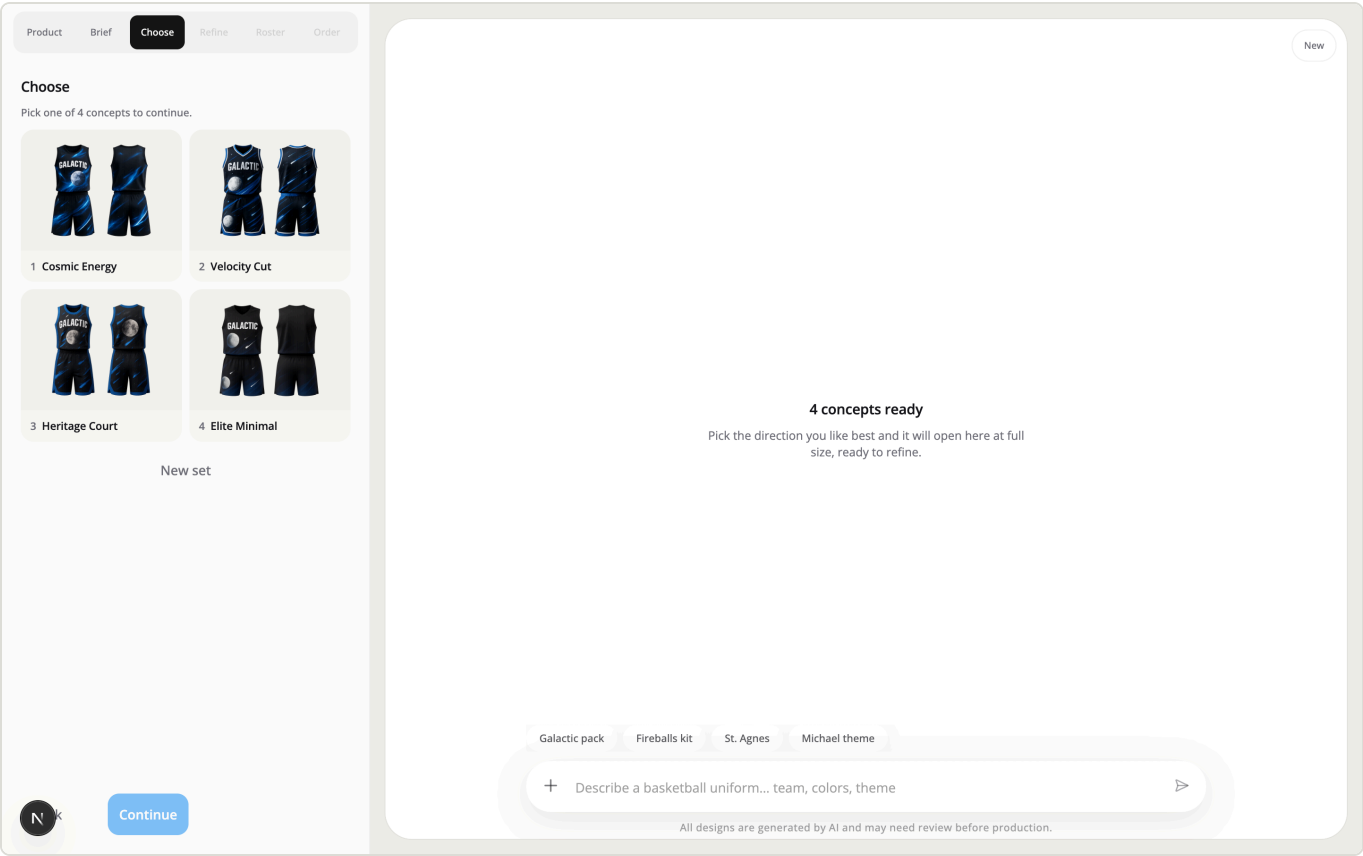
04-desktop-concepts-no-selection-locked.png Downstream steps are locked while no concept is selected. The tabs are disabled and marked aria-disabled="true" for assistive technology.

5. Live production output

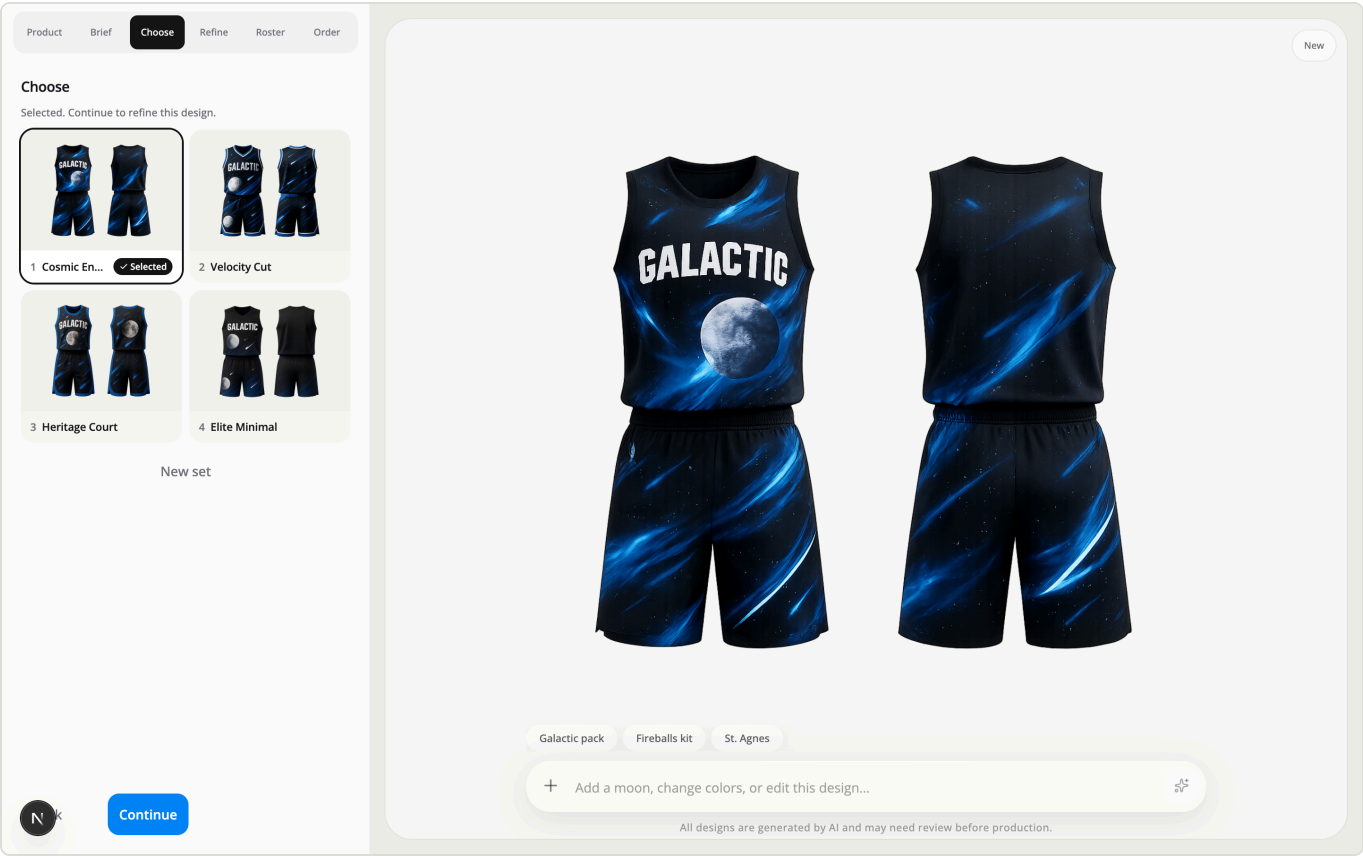
The whole customer journey was then run a second time against the **live production image model** on the GALACTIC brief, driving the real application in a real browser exactly as the mocked run did. **All 32 checks passed with 0 failures**, including the responsive sweep and the Shopify handoff capture. The screenshots in this section are real model output and show what the customer actually receives.



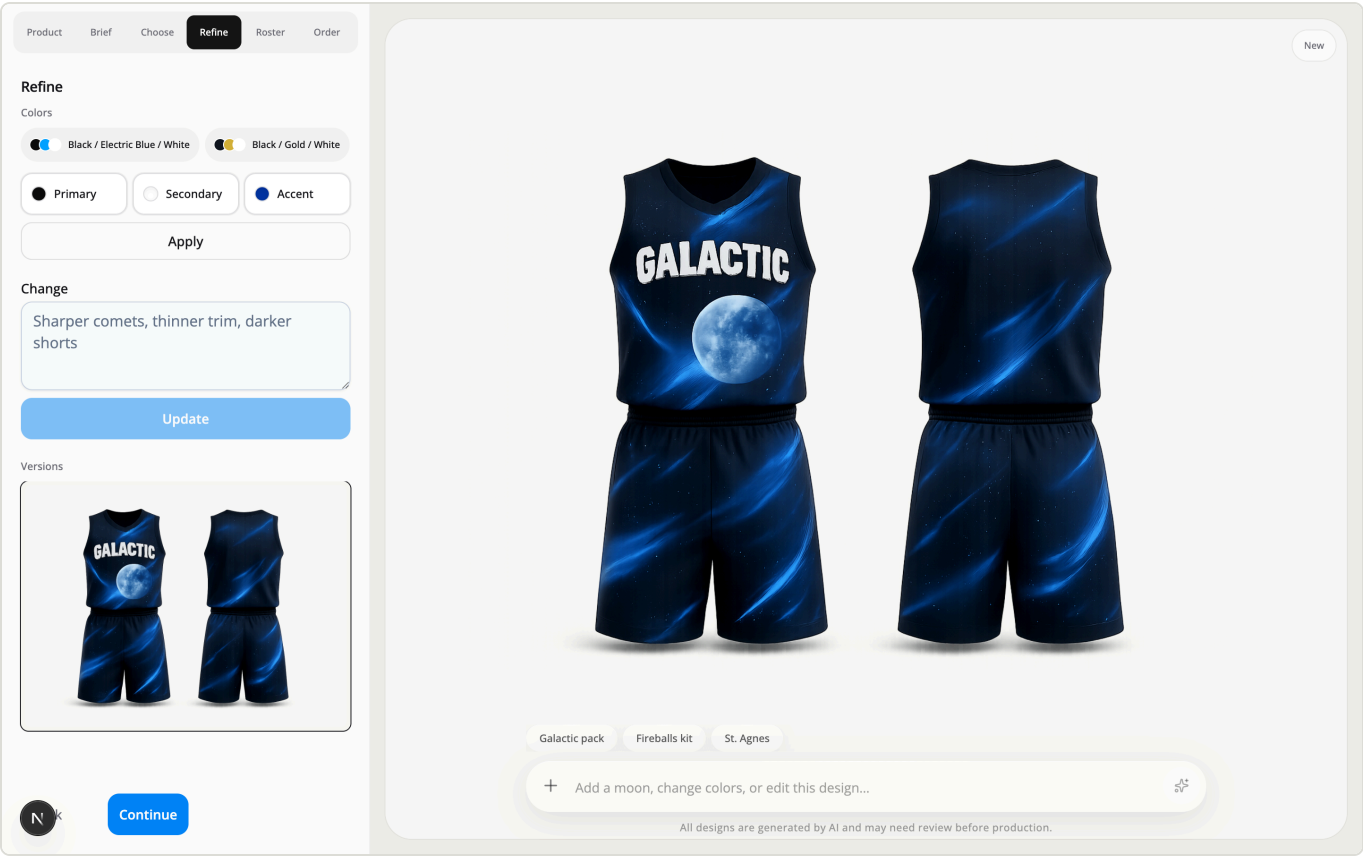
live-03a-desktop-four-concepts-detail.png The four live-model concepts from the GALACTIC brief. One brief, four genuinely different finished uniforms — a flowing comet sweep, sharp angular speed panels, a structured retro-court treatment, and a restrained minimal kit — each with the team wordmark correctly spelled on the chest and a clean numbering area on the back.



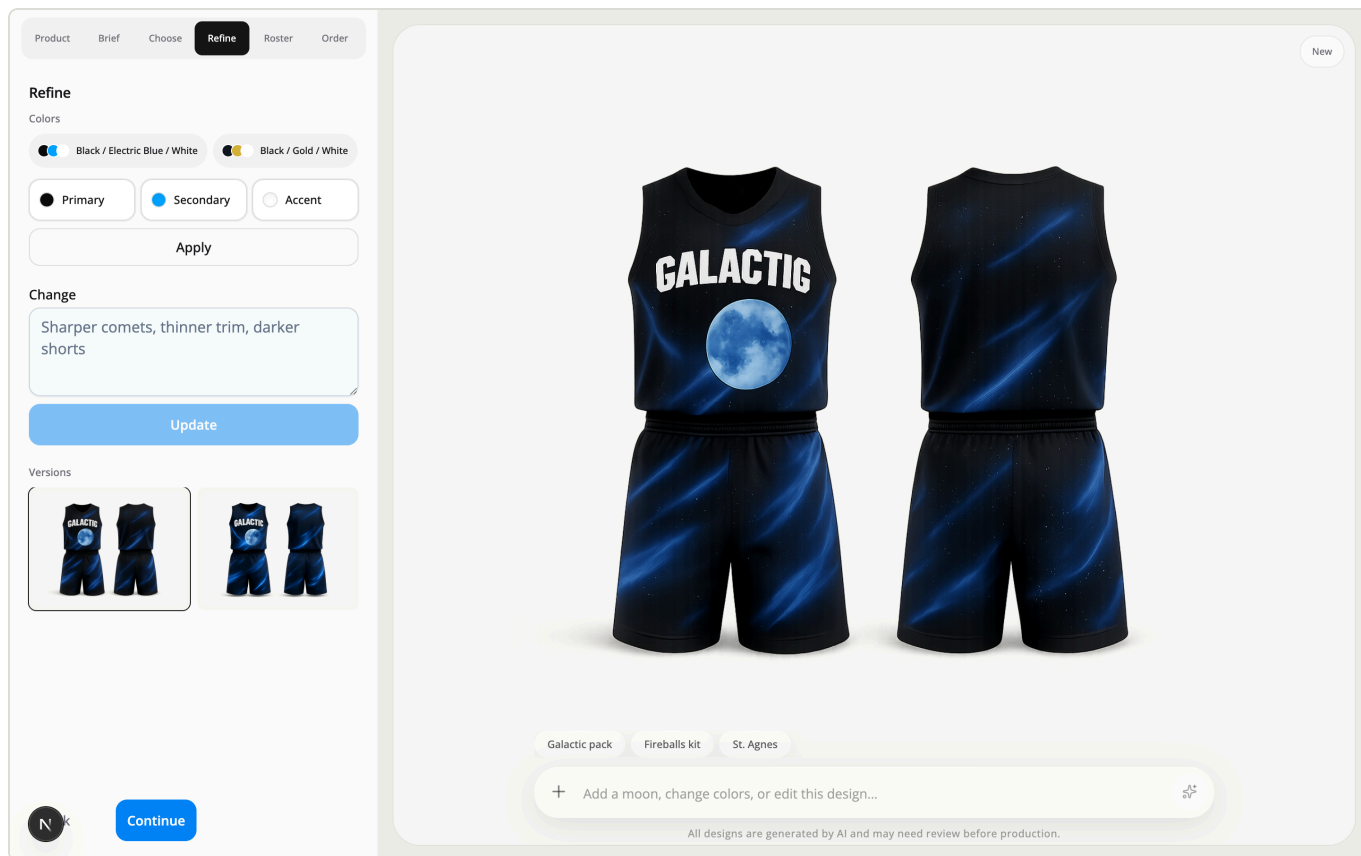
live-03-desktop-four-concepts.png The live-model concepts in the Choose step. Nothing is selected, Refine / Roster / Order are dimmed, and the canvas asks the customer to choose.



live-05-desktop-selected-concept.png A live-model concept selected. The chosen render fills the preview at full size and the downstream steps unlock.



live-06-desktop-refinement.png Live refinement. The instruction sharpened the comet trails and adjusted the trim while keeping the same design identity — the moon, the cosmic sweep and the garment structure all carry over, and the previous version is retained in the version strip.



live-07-desktop-color-variation.png Live colour variation. The palette is re-applied to the same design rather than generating a new one, and the garment cut, motif placement and composition are preserved. This capture also shows the open wordmark issue described below — the recolour has rendered the final letter incorrectly.

A second live run was also captured on a black-and-orange flame brief, shown below, to confirm the four directions behave consistently across different briefs and colour systems rather than only on the cosmic theme.

Each render is a finished, wearable uniform visualisation, exactly as the generation contract requires:

- Sleeveless basketball jersey with matching shorts, on a clean neutral background.
- Front and back presentation side by side in a single render.
- Realistic sportswear construction — visible fabric, panel seams, ribbed neckline and armhole trims, contrast binding.
- Team wordmark and number on the front chest, with a large clean number on the back.
- No models, mannequins, hangers, backgrounds, watermarks or invented sponsor marks.

Choose

Selected. Continue to refine this design.



1 Cosmic En... ☒ Selected



2 Velocity Cut



3 Heritage Court

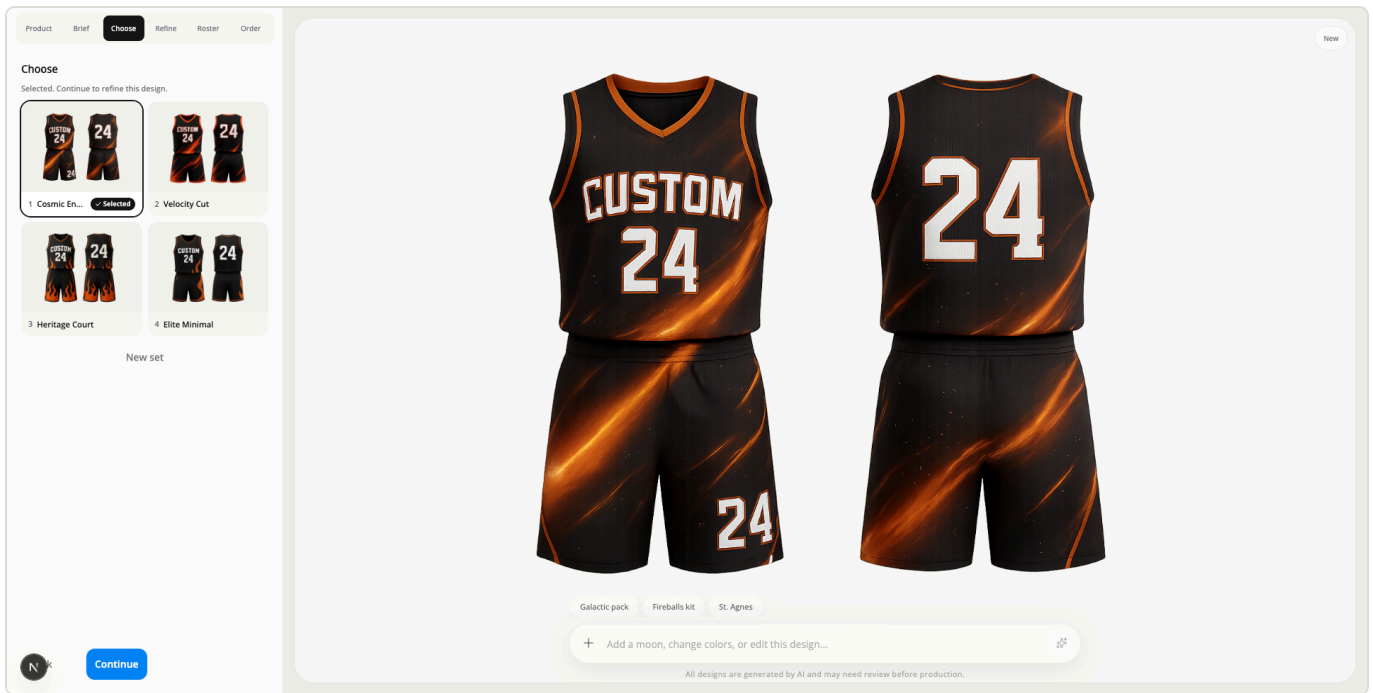


4 Elite Minimal

New set

26-live-ai-concept-choice-grid.png The four live-model concepts in the choice grid. One brief, four genuinely different finished uniforms: a flowing comet sweep, sharp multi-blade cuts, full traditional flames, and a restrained single-flame minimal treatment.

The four directions are clearly distinct rather than four near-duplicates of one design, which is the entire commercial point of offering a choice. The pages that follow show each direction selected in turn, at full preview size.



22-live-ai-concept-1-cosmic-energy.png Direction 1 — Cosmic Energy, selected. A sweeping comet-trail graphic across the jersey and shorts. The card is badged "Selected" and the chosen render fills the main preview.



23-live-ai-concept-2-velocity-cut.png Direction 2 — Velocity Cut, selected. Sharp multi-blade speed forms with contrast neckline and armhole trims.



24-live-ai-concept-3-heritage-court.png Direction 3 — Heritage Court, selected. A traditional full-flame treatment with a classic block wordmark and collegiate proportions.



25-live-ai-concept-4-elite-minimal.png Direction 4 — Elite Minimal, selected. A restrained single-flame accent with clean negative space and precise trim geometry.

What these confirm on live output: four independent renders from one brief, all four presented together in the Choose step, explicit selection promoting one render to the full preview, selection-scoped refinement and recolouring, the front/back wearable-uniform contract, and legible team wordmark and number placement.

Switching the selected direction updates the main preview immediately, with no trace of the previously viewed concept.

Two output defects found on live output and corrected

Running against the live model surfaced two problems that the preview renderer could not expose, because both are behaviours of the image model rather than of the application. Both were corrected in the image prompt and re-verified on fresh live renders.

ISSUE	WHAT HAPPENED	CORRECTION
Third-party brand marks	On some renders the model added a sportswear manufacturer mark to an otherwise blank chest or shorts panel, which is not acceptable on a customer's uniform.	The prompt now carries an explicit, absolute prohibition on any manufacturer or apparel-brand mark, naming the marks that actually appeared, positioned as the final instruction where the model weights it most heavily. Fresh renders came back clean.
Team-name spelling drift	The chest wordmark occasionally lost or substituted a letter — <i>GALACTIE</i> on first generation, and <i>CALACTIC</i> after a recolour, because editing makes the model redraw the lettering.	The wordmark is now spelled out character by character with its letter count stated as a checkable constraint, and every edit mode instructs the model to carry the existing wordmark over untouched rather than re-letter it. Generation is now reliable — all four concepts spell <i>GALACTIC</i> correctly. Recolouring can still drift a single letter; see below.

Open item, stated plainly Text rendering is the one genuinely probabilistic part of image generation. On the generation path the fix holds: every concept in the verification runs spelled the team name correctly, and no brand marks appeared. On the *recolour* path a single letter can still drift, because the model redraws the lettering when it edits the image, and no prompt wording fully prevents that.

The durable fix is to stop asking the image model to draw the team name at all — generate the garment without the wordmark and composite the text deterministically from the team-name field, so it is always pixel-correct and always matches what the customer typed. That is a contained, well-understood change and the recommended next step. Until it is in place, a design should get a glance at the wordmark before production, which is also why the interface carries the standing note that designs are AI-generated and may need review before production.

6. Root cause of the workflow defect

This section is included so the change is auditable. It is a technical note, not a criticism of any contributor.

Before

The repository already contained a complete four-concept generator with the four art directions. However, the production generation hook that both customer entry points called was wired to an older single-render helper. Its effect was to generate one render, store it as a single concept, select that concept automatically, and jump the customer past the Choose step. Because the concept was auto-selected, the downstream gating never engaged, so the flow appeared to work while the concept-selection experience was absent.

Why it was not caught

The existing verification scripts asserted that the four-concept generator *existed in the source files*. They could not detect that the production path never called it. The same scripts simultaneously asserted a "three labeled designs" collage prompt, which was inconsistent with a four-selectable-concept product.

After

There is now exactly one fresh-generation entry point. It runs the four-direction generator, stores four concepts, deliberately does *not* select one, and routes the customer to the Choose step. The single-collage helper and the conflicting collage prompt mode were removed so the path cannot be reintroduced by accident. The verification suite was rewritten to execute the real store, generation client and generation hook against a stubbed network, and it now fails if the production flow stores one concept, auto-selects, or skips the Choose step.

A collage image cannot satisfy the commercial requirement: a single image containing several designs cannot be individually selected, refined, exported or attached to an order line. Each concept must be its own render, which is what the flow now produces.

7. AI generation contract

Every concept is generated as a **finished, wearable uniform visualisation** — not a print graphic to be applied later. The image prompt requires:

- A sleeveless basketball jersey with matching basketball shorts.
- Front and back presentations in the same image, clearly separated.
- Realistic sportswear construction: believable fabric, cut-and-sew detail, crisp trims, clean seams.
- The exact requested team name on the front chest.
- A clean back area reserved for the player name and number.

The prompt explicitly excludes flat sublimation textures, UV maps, pattern sheets, fabric swatches, technical templates, isolated artwork, people, mannequins, hangers, stadiums, watermarks, invented sponsor marks and

unrelated branding. Renders are produced in landscape so the front and back presentations both have room.

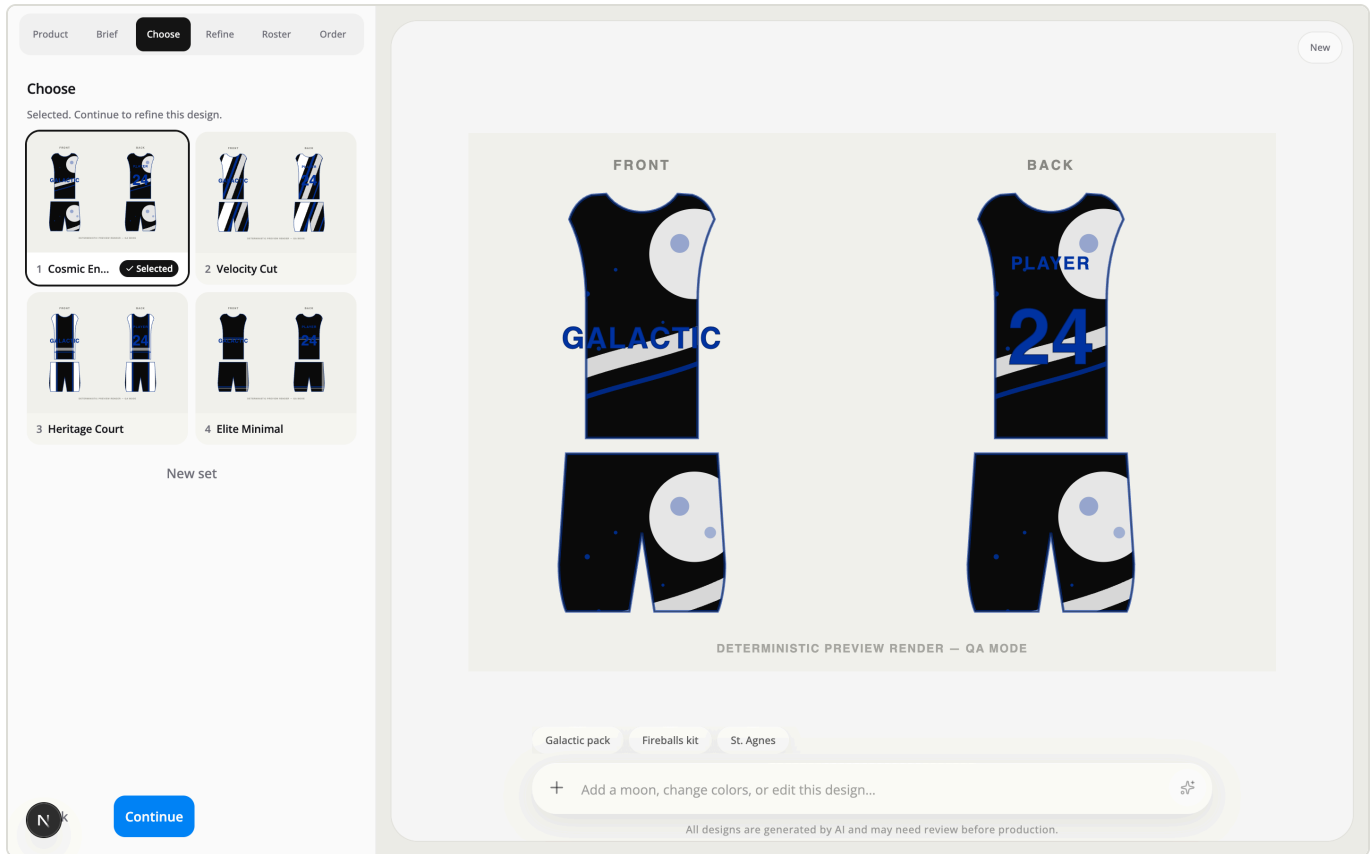
Concept directions

DIRECTION	ART DIRECTION SUPPLIED TO THE IMAGE MODEL
Cosmic Energy	Bold galactic graphics, sweeping nebula motion, comet trails, star fields and dramatic angular panels.
Velocity Cut	Fast court aesthetic with sharp diagonal cuts, speed lines and layered geometric panels.
Heritage Court	Retro-modern identity with structured side panels, vintage court geometry and classic proportions.
Elite Minimal	Luxury minimal kit with clean negative space, precise trim geometry and premium tonal panels.

Each request receives the customer's brief plus one of these directions and an instruction to make the composition clearly different from the other proposed directions. This is what produces four genuinely different options from a single brief rather than four near-duplicates.

8. Concept selection

Selecting a concept card promotes exactly that concept: the main preview switches to its render, its design ID becomes the working design ID, its palette is adopted, and version history is cleared so nothing from a previously viewed concept can leak forward. The card is marked with a visible "Selected" state and `aria-pressed="true"`.

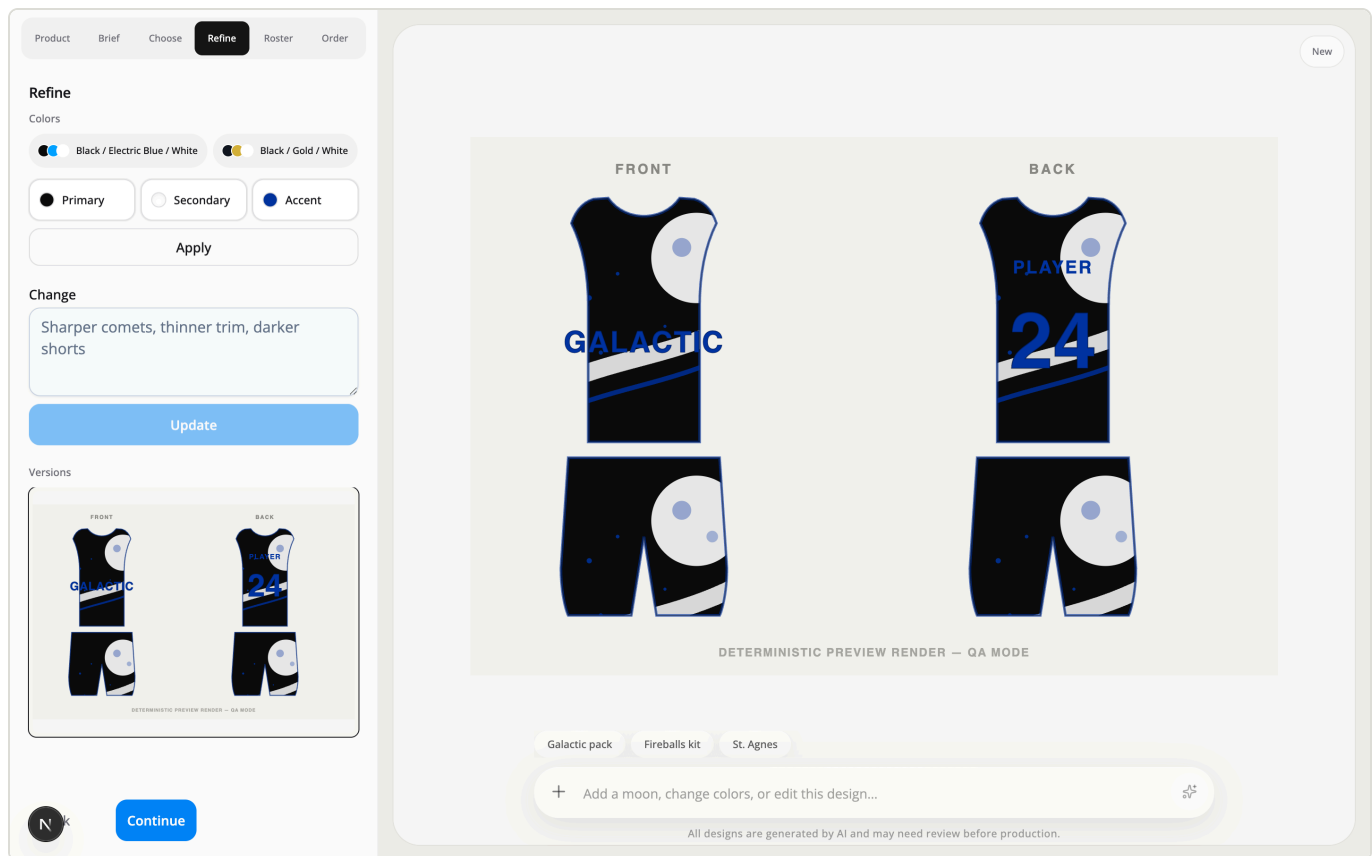


05–desktop–selected–concept.png A concept has been selected. The main canvas shows the chosen uniform, the card is badged "Selected", and Refine, Roster and Order are now available.

Verified: Cosmic Energy selected, preview + designId updated, downstream unlocked. Switching between concepts was also verified in both directions: selection switches both directions without leaking artwork or history.

9. Refinement of the selected design

Refinement is an *edit* of the selected render, not a new generation. The current render is supplied to the image model as the edit source and the request is sent in refine mode, with an instruction to keep every unmentioned part of the uniform and its presentation consistent. The selected concept and its design ID are unchanged by a refinement, and each refined render is added to a version history the customer can step back through.

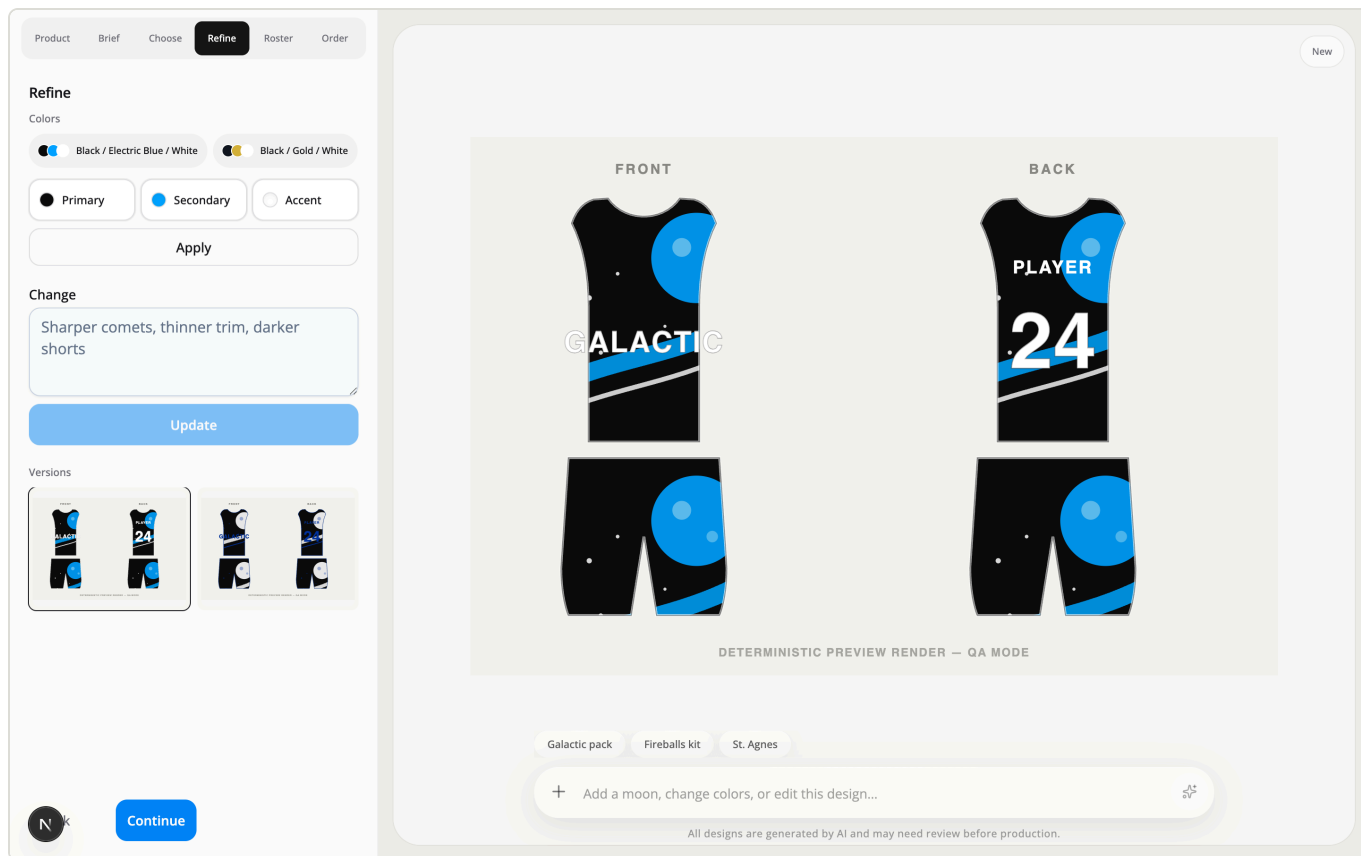


06-desktop-refinement.png The Refine step after applying the instruction "Add a moon detail near the lower jersey and sharpen the comet trails." The refined render replaces the preview and is recorded as a version.

Verified: history 1, designId preserved, 5 total generation calls. A follow-up prompt typed into the studio bar after a concept is chosen is treated as a refinement rather than a new set, so a customer cannot lose their chosen design by describing a tweak.

10. Colour variation

Colour variation is handled separately from text refinement. It sends the current render as the edit source with an instruction to preserve the garment cut, front/back presentation, graphic composition, motifs, panel layout and trims, and to change only the palette.



07-desktop-color-variation.png The black / electric blue / white palette applied to the selected concept. The garment structure and motif placement are retained; only the colours change.

Verified: palette #0A0A0A / #00A3FF / #FFFFFF, garment identity preserved.

11. Team name and logo handling

The team name is captured as an explicit field and injected into the image prompt as an exact string, so the front chest wordmark reflects what the customer typed rather than an invented name. The prompt also forbids additional typography, slogans and invented sponsor marks. In the QA run the team name `GALACTIC` was carried from the brief through generation, selection, refinement, roster, export and the Shopify payload.

Logo upload accepts PNG, JPEG and WebP and rejects other types. An uploaded logo is stored and referenced by URL. When a logo has been provided, the image prompt reserves a crest position on the front chest but does not invent a logo mark.

Deliberate constraint: logo artwork is never embedded as inline image data in the Shopify line-item properties. Only external URLs are passed. This was asserted in both the behavioural test suite and the live browser payload capture.

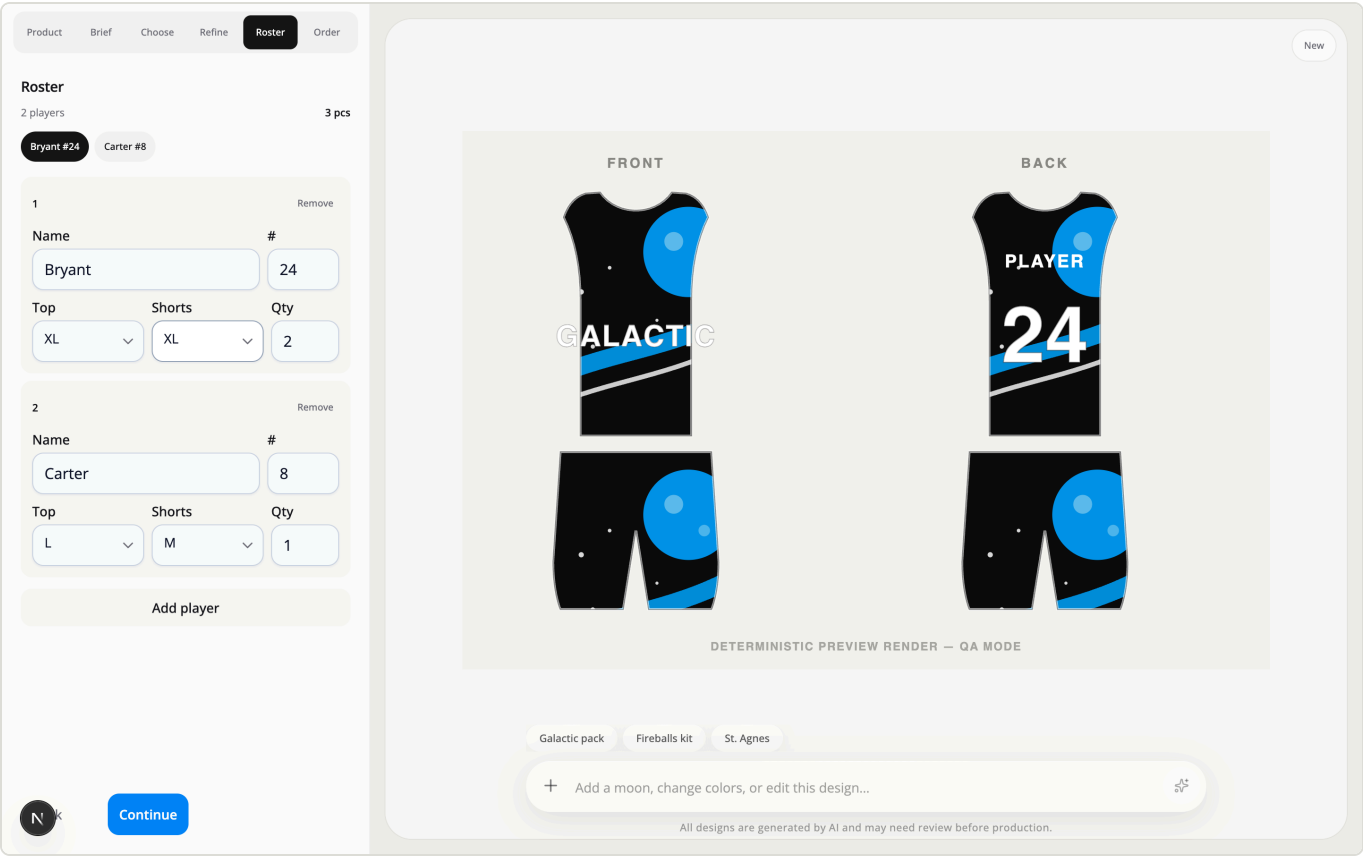
12. Product flow

The catalogue offers 18 products, default "Basketball Uniform" across basketball, soccer, volleyball, baseball, flag football, track and training, including coordinated uniforms as well as standalone jerseys and shorts. Changing the product resets concepts, selection, artwork and design ID, which prevents a design generated for one garment from being ordered as another. Prompt suggestions shown above the studio prompt bar are filtered to the selected product.

Verified: jersey <-> uniform switch clears concepts.

13. Roster

Roster rows are independent records. Each row captures a player name, number, top size, shorts size and quantity. Sizes are chosen from the mapped production size range, duplicate numbers are rejected, and removing a row affects only that row.



08–desktop–roster.png The roster step with two independent players: Bryant #24 at XL top / XL shorts, quantity 2, and a second player with different sizes and quantity.

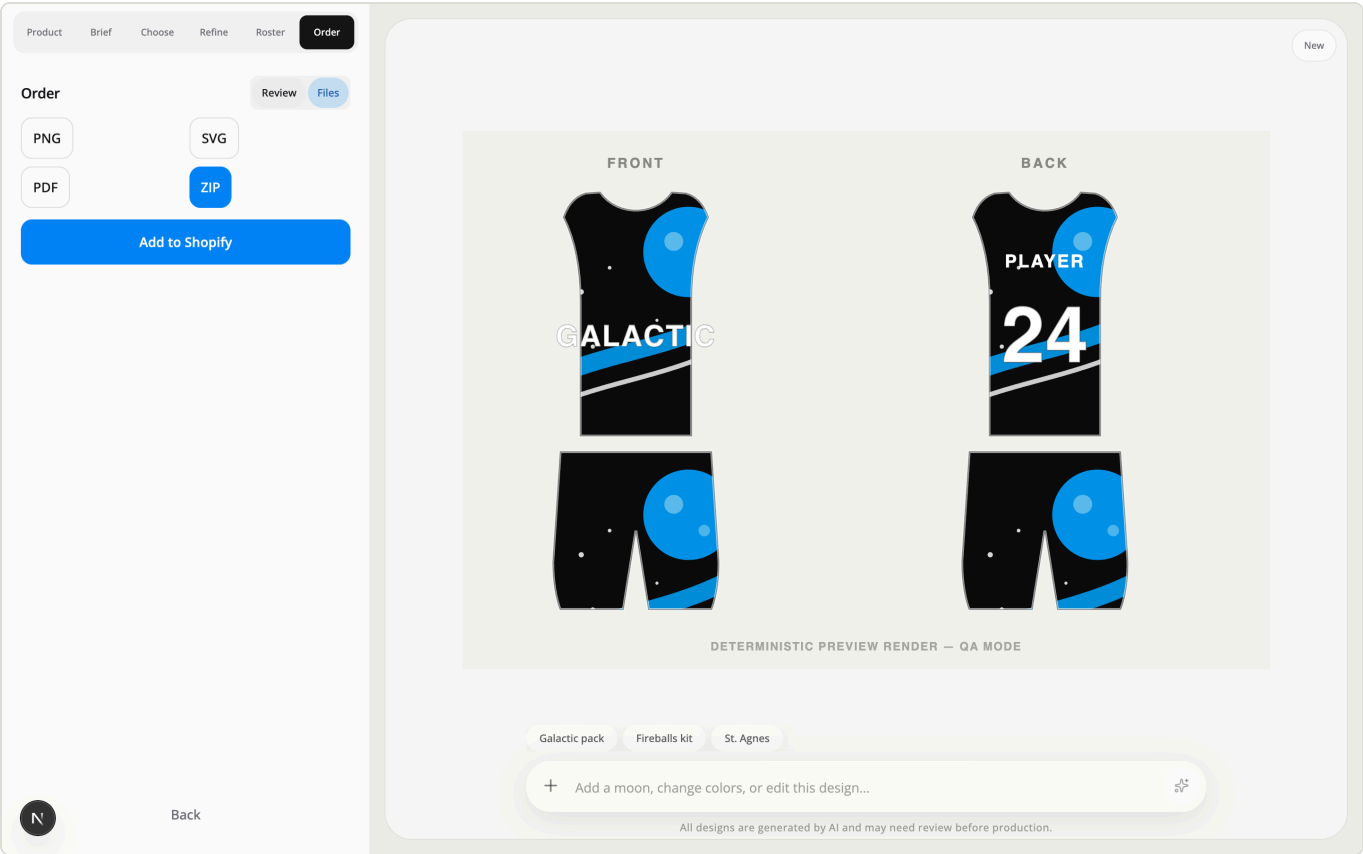
Verified: Bryant #24 XL/XL x2, Carter #8 L/M x1. Deletion was verified separately: deleted one row, roster now 2.

14. Production exports

Four output formats are offered, all generated from the current preview so they always reflect the selected and refined design rather than an earlier concept.

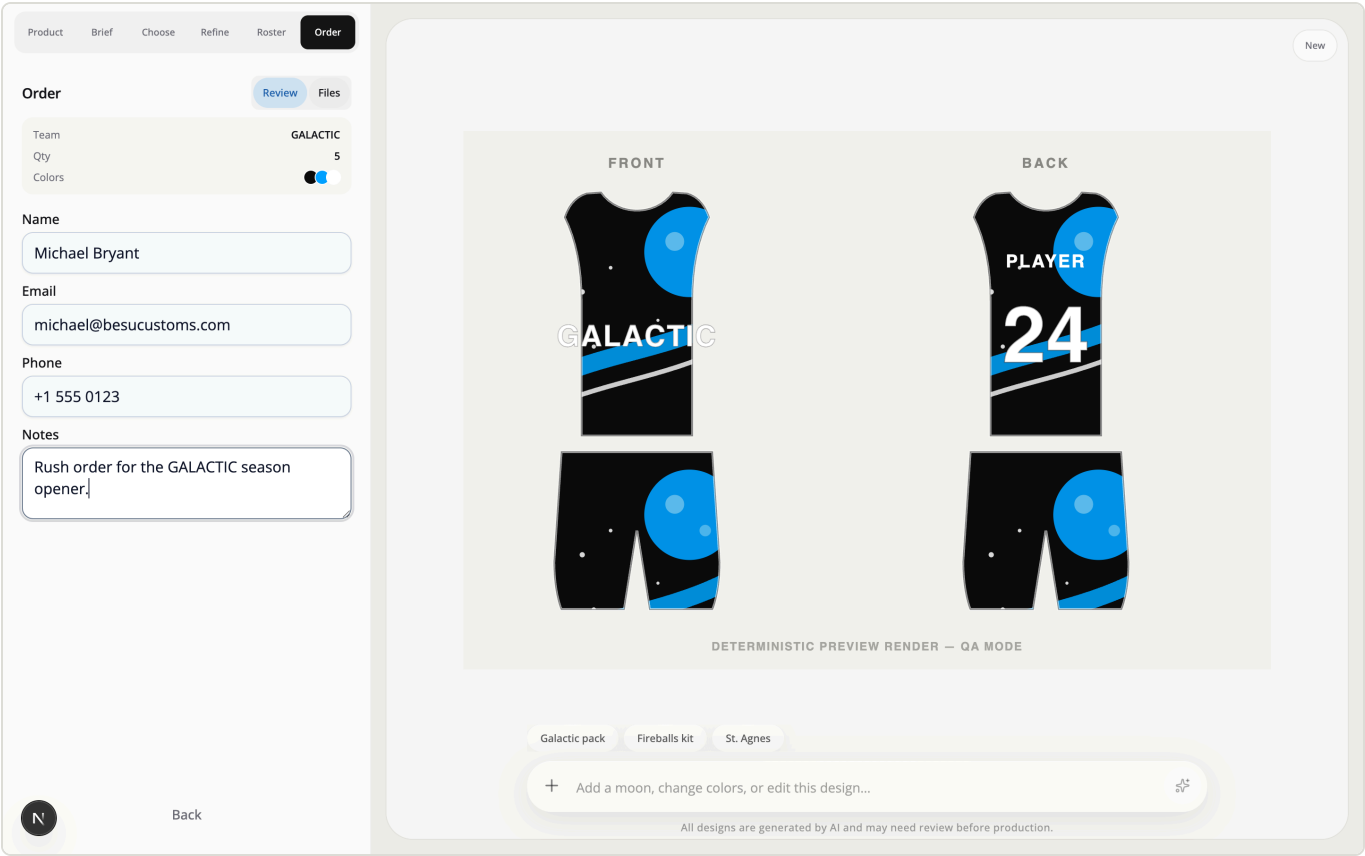
FORMAT	CONTENTS
PNG	High-resolution raster of the front and back presentation, 2400 px wide at the render's true proportions.
SVG	The preview document with the AI render embedded. This is a raster image wrapped in SVG, not editable vector uniform artwork. The file carries a comment stating this.
PDF	Production summary — design ID, garment, sport, style, team, palette, customer, total quantity, artwork URLs, the full roster and production notes — plus a page with the front and back renders.
ZIP	Front and back SVG, front and back PNG, the production PDF, and an order JSON containing the design ID, garment, team, palette, artwork URLs, roster, customer details and export timestamp.

Stated accurately: the designer produces direct raster AI renders. The SVG export is therefore a raster image inside an SVG wrapper. It is suitable for placement and production reference, but it is not a redrawable vector uniform file, and it is not described as one anywhere in the product or in this report.



10-desktop-export-options.png The production file options, enabled once a design is selected.

A defect was found and fixed here. The raster export previously drew the landscape render onto a fixed portrait canvas, which stretched the uniform vertically in the PNG, PDF and ZIP output. Exports now derive their height from the render's own proportions. Verified in the browser: 2400x1600 raster from the current render.



09-desktop-order-review.png The order review step with customer details, team, total pieces and palette.

15. Shopify order handoff

The designer runs inside the Shopify storefront and hands the order to the parent page by posting a `besu:checkout` message. To verify this genuinely rather than by inspection, the QA harness loaded the designer inside a parent frame on the same origin, completed the order in the frame, and captured the posted message.

Captured payload

FIELD	CAPTURED VALUE
Message type	<code>besu:checkout</code>
Design ID	<code>b08e6d57-e4bd-47a9-a4d5-37a582d6e558</code>
Front artwork	External asset URL (no inline image data)
Line items	2
Mapped Shopify variant IDs	<code>51354887192854</code> , <code>51354887127318</code>
Quantities	[2,3]
Players on line items	Bryant#24, Carter#8
Design ID on every line item	Yes
Inline base64 image data	None

Every roster row becomes its own line item with its own mapped Shopify variant, quantity, player name and number, while the shared design metadata (design ID, team, palette, artwork URLs, customer details) is attached to each item. A size that is not mapped to a Shopify variant blocks checkout rather than sending an unfulfillable order, and two different roster configurations produce two distinct line items rather than merging.

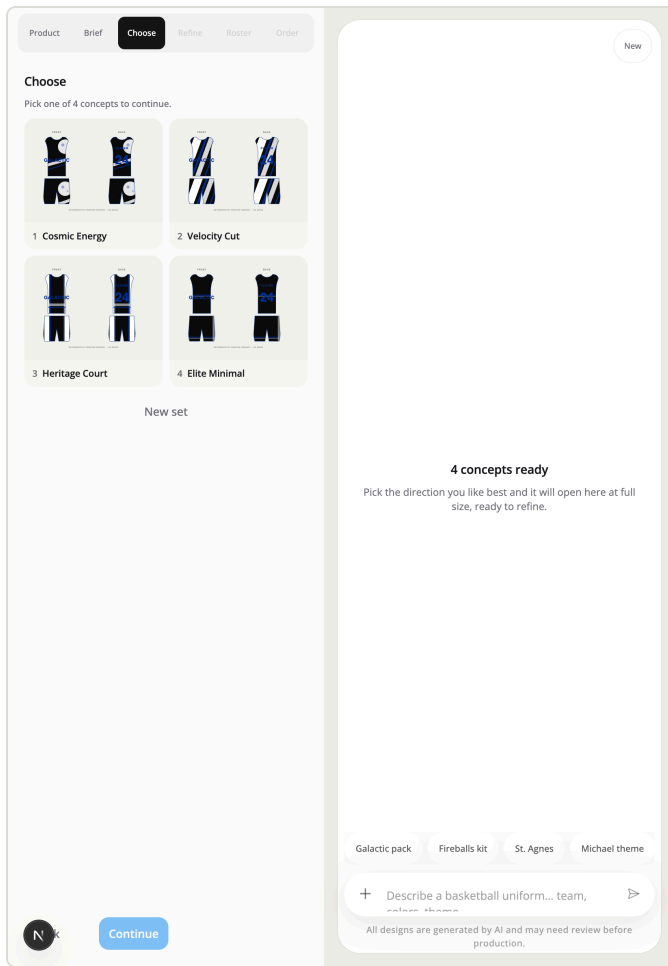
Scope **What was and was not tested.** The payload above was captured from a real browser, from a real parent frame, using the real production code path. **A live order was not placed against the production Shopify store** — that requires the storefront environment and its credentials, which are outside this verification. The remaining step for go-live is to confirm the storefront listener consumes this payload and creates the cart as expected.

16. Responsive QA

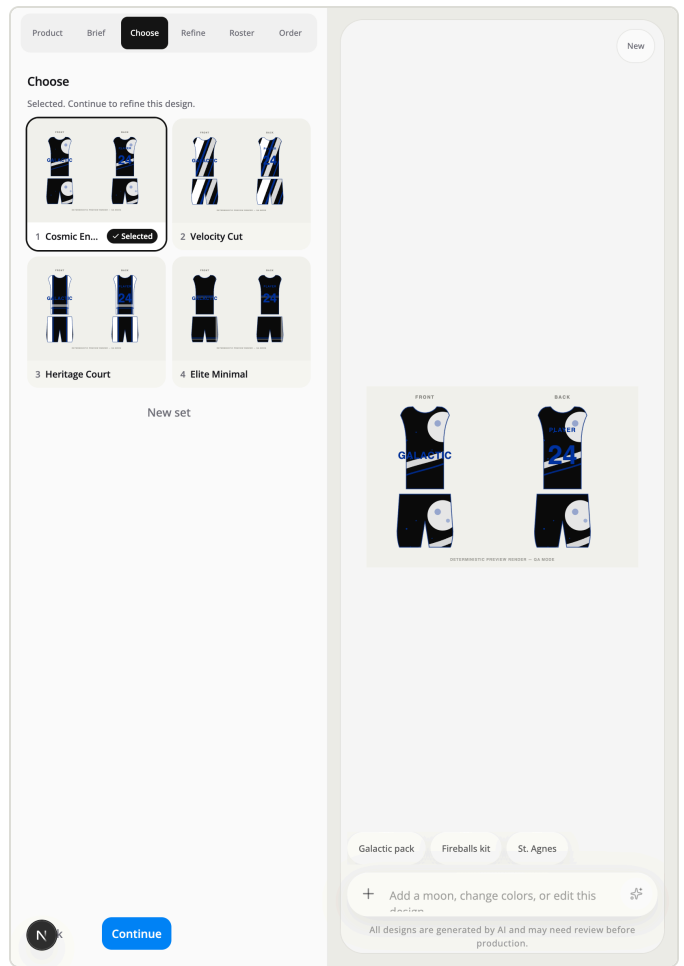
Every viewport below was tested against the live application on the Choose step, checking that all four concepts render, that there is no horizontal overflow, that the prompt bar is reachable inside the viewport, that the garment preview is not collapsed, that concept cards remain readable, and that no interactive control falls below a 32 px touch target.

VIEWPORT	CONCEPTS	H-OVERFLOW	PROMPT BAR	CARDS READABLE	TOUCH TARGETS
Desktop 1440x900	4	None	Yes	Yes	Pass
Laptop 1280x800	4	None	Yes	Yes	Pass
Tablet portrait 820x1180	4	None	Yes	Yes	Pass
Tablet landscape 1180x820	4	None	Yes	Yes	Pass
Mobile 390x844	4	None	Yes	Yes	Pass
Small mobile 360x800	4	None	Yes	Yes	Pass
Mobile 375 375x812	4	None	Yes	Yes	Pass

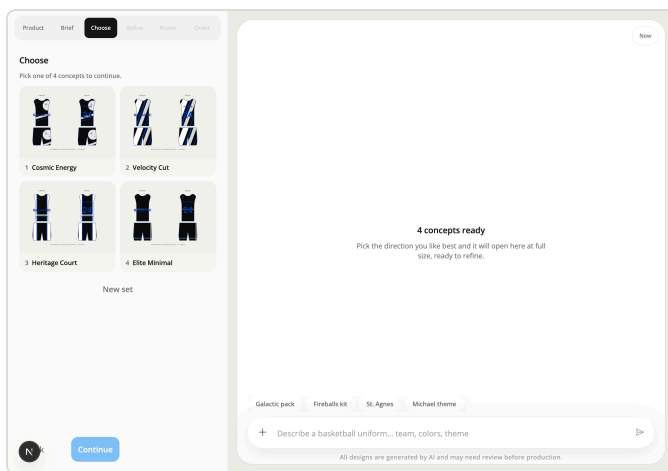
A defect was found and fixed here as well: at 360 px the step navigation overflowed the page by 16 px because the horizontally scrolling tab strip was sized to its own content. The scroll container and the content row are now separate elements, so the tabs scroll instead of pushing the layout wider.



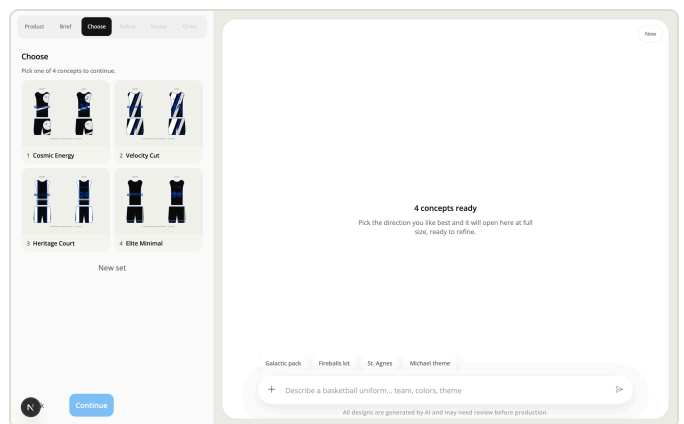
11-tablet-four-concepts.png Tablet portrait, 820 × 1180 — Choose step.



12-tablet-selected-concept.png Tablet portrait — concept selected.

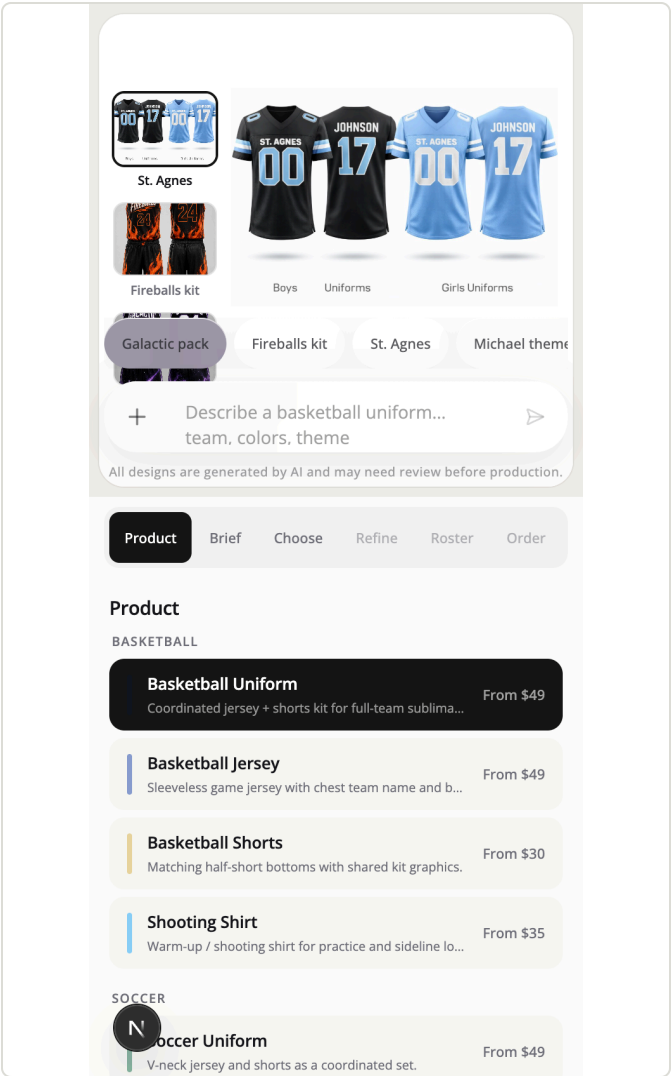


20-tablet-landscape-four-concepts.png Tablet landscape, 1180 × 820.

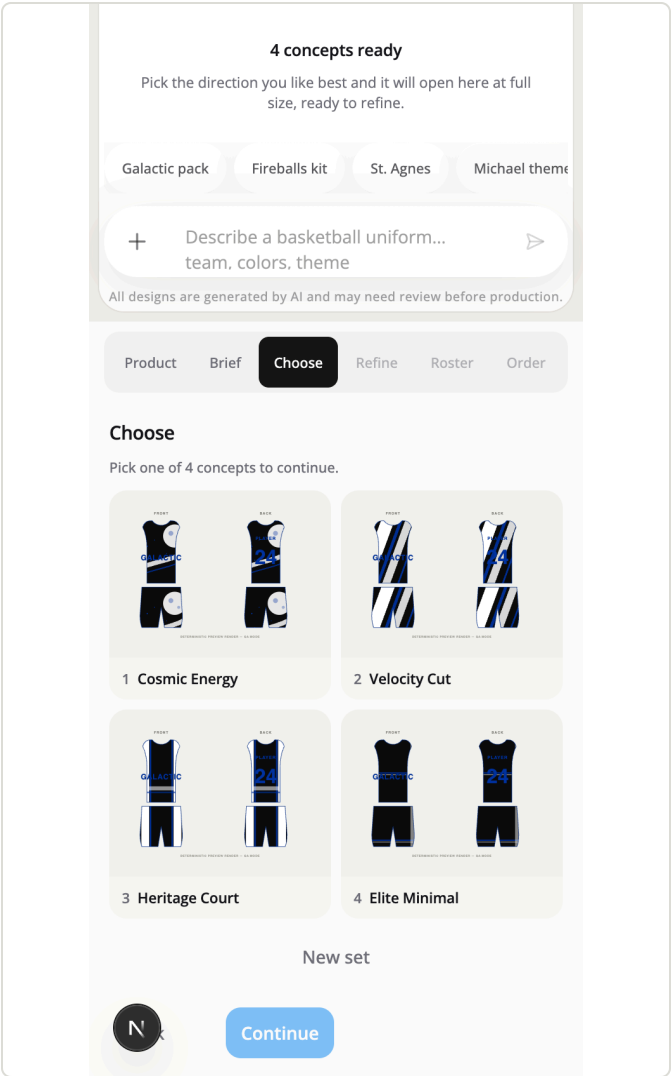


19-laptop-four-concepts.png Laptop, 1280 × 800.

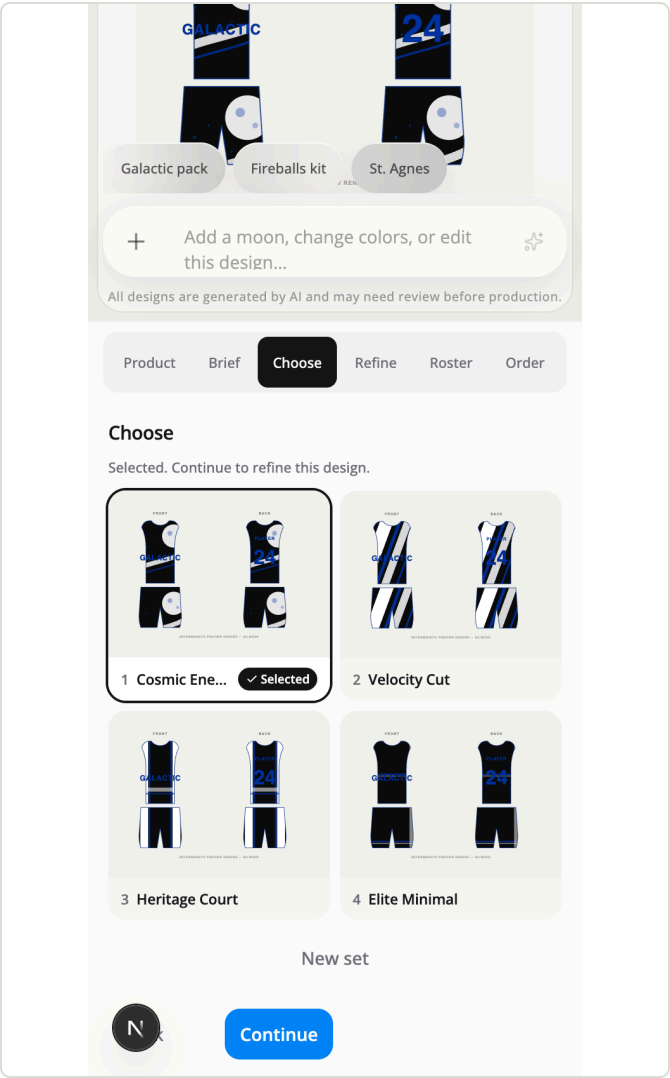
Mobile



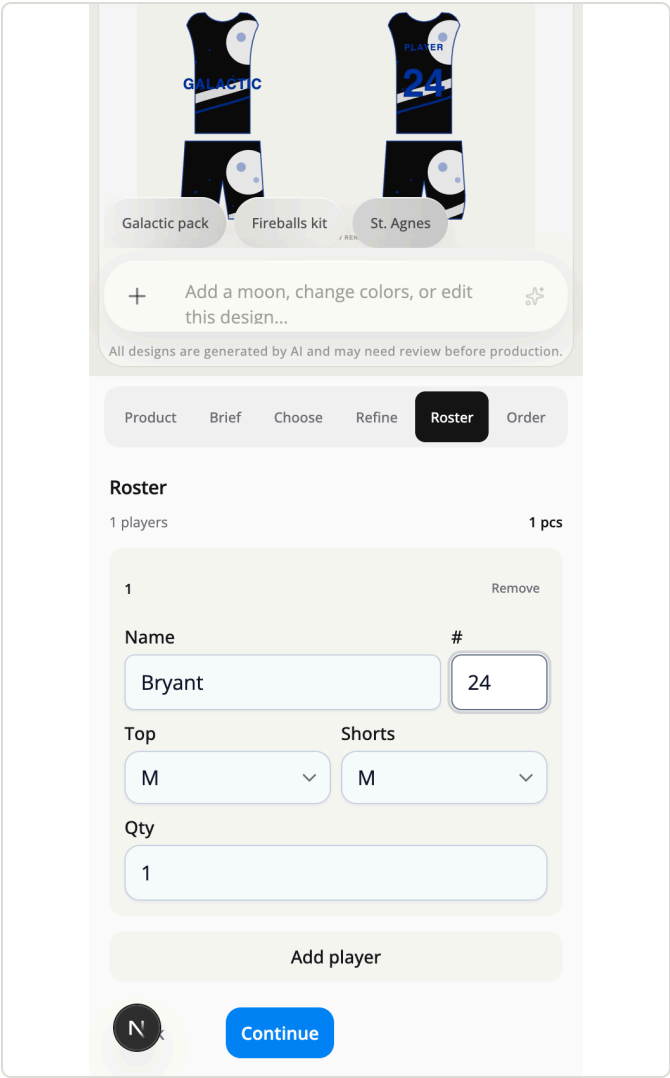
13-mobile-initial.png Mobile 390 × 844 — first load.



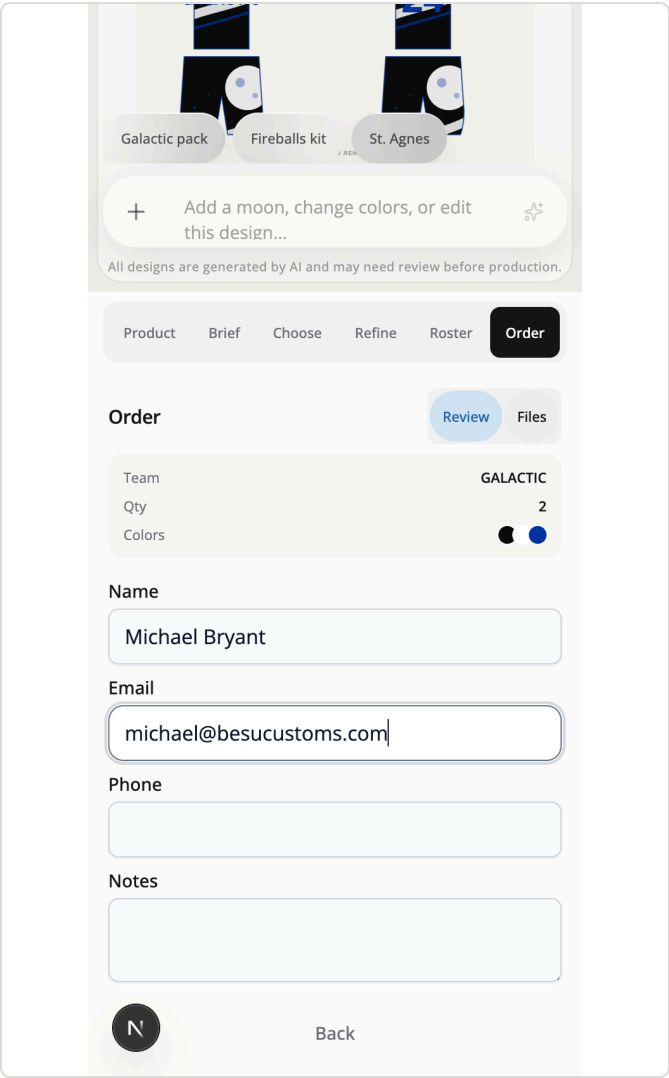
14-mobile-four-concepts.png Mobile — four concepts.



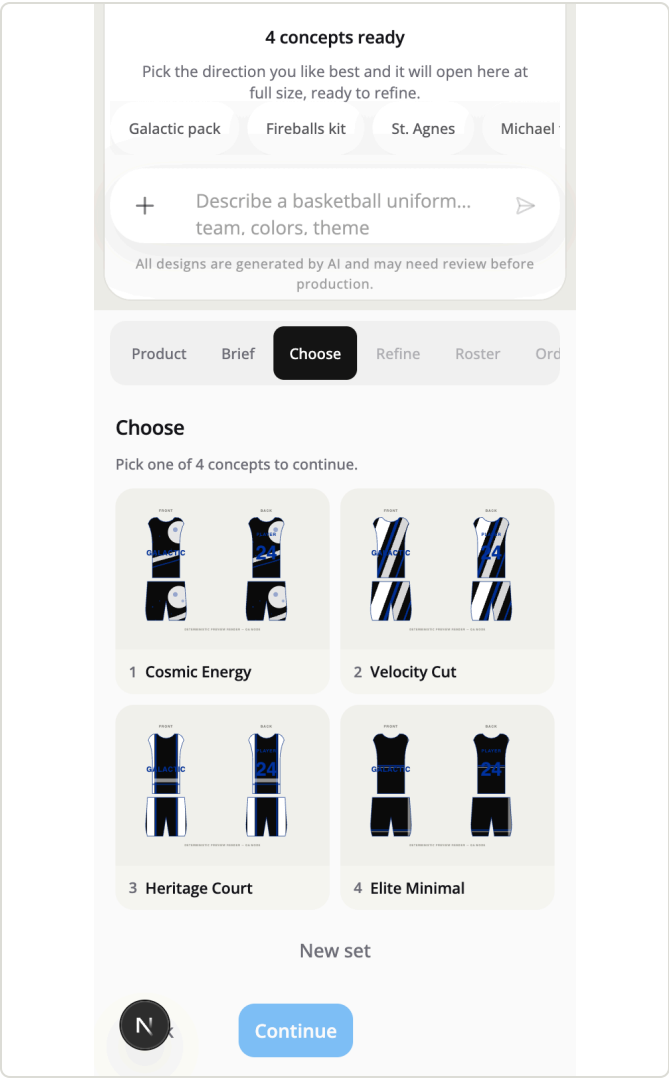
15-mobile-selected-concept.png Mobile — concept selected.



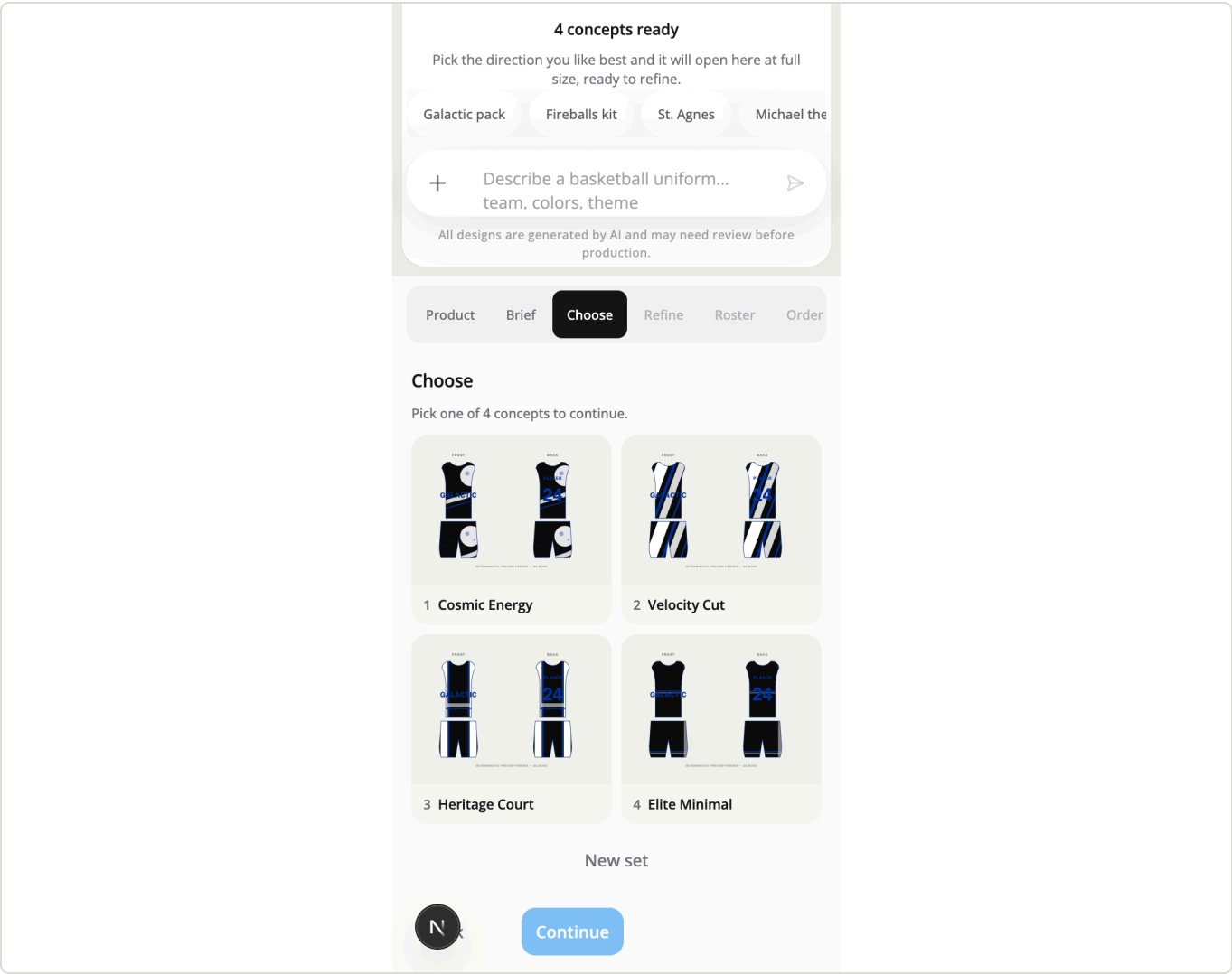
16-mobile-roster.png Mobile — roster entry.



17-mobile-order.png Mobile — order review.



21-mobile-small-four-concepts.png Small mobile 360 × 800 — after the overflow fix.



18-mobile-375-four-concepts.png Mobile 375 px width — Choose step.

17. Accessibility and UX checks

These are release-blocker checks, not a full accessibility audit.

CHECK	RESULT	DETAIL
Keyboard focus reaches concept cards and the prompt input	Verified	19 focusable controls, prompt focusable
All visible buttons have an accessible name	Verified	15 buttons and 5 inputs all labelled
All visible images have alt text	Verified	15 buttons and 5 inputs all labelled
All visible inputs are labelled	Verified	15 buttons and 5 inputs all labelled
Reduced-motion preference honoured	Verified	shell renders with prefers-reduced-motion: reduce
Locked steps announced to assistive technology	Verified	Locked tabs set <code>aria-disabled="true"</code> and are not focusable.
Concept cards expose selection state	Verified	Cards use <code>aria-pressed</code> ; the selected card also carries a visible badge.
Touch target size	Verified	No interactive control under 32 px tall at any tested viewport.

18. Console and network health

CHECK	RESULT	DETAIL
Uncaught page exceptions	None	Clean across the whole run
Console errors	None	Clean across the whole run
Failed network requests	None	No 4xx/5xx responses and no failed loads
Duplicate generation requests	None	A concurrent submit is refused by a single in-flight guard; 10 generation requests were issued across the entire QA run.

Generation is rate limited server side to a bounded number of requests per minute per IP. One fresh concept set costs four requests. Both customer entry points share one in-flight guard, so a repeated submit or a React re-render cannot issue a duplicate set.

19. Automated verification

All four project command suites were run on the delivered build.

COMMAND	RESULT	OUTPUT SUMMARY
<code>npm run typecheck</code>	Pass	tsc --noEmit completed with no type errors.
<code>npm run lint</code>	Pass	ESLint over the designer app, components, lib and API routes with --max-warnings=0: no errors, no warnings.
<code>npm run test:designer</code>	Pass	3 verification suites passed: 25 named assertions (0 failed) plus the Shopify checkout payload generator, which asserts by throwing on any invalid payload.
<code>npm run build</code>	Pass	Next.js production build compiled successfully and generated the full route manifest.

Behavioural test coverage

The verification suite was rewritten so it proves runtime behaviour rather than the presence of source text. It loads the real designer store, the real generation client and the real generation hook, stubs the network, and then asserts the flow. It fails if:

- the four-concept generator exists but the production hook does not call it;
- a fresh generation stores fewer than four concepts, or four concepts that share a render or design ID;
- a fresh generation auto-selects a concept, sets artwork or a design ID, or navigates past the Choose step;
- refinement is possible without a selection, or does not supply the previous render as the edit source;
- refinement or colour variation changes the selected concept or its design ID;
- a concurrent submit issues a duplicate set of requests;
- the checkout payload loses the selected design ID, carries inline image data, or contains an unmapped variant;
- the retired single-collage generator or its "three labeled designs" prompt reappears.

20. Browser QA matrix

CHECK	RESULT	DETAIL
designer shell renders on the public route	PASS	v17-studio-minimal
product step lists production products and basketball uniform is selected	PASS	18 products, default "Basketball Uniform"
product change resets any prior design state	PASS	jersey <-> uniform switch clears concepts
brief step accepts the team name and design brief	PASS	team GALACTIC, 217 char brief

generating produces four separate AI concept requests	PASS	4 requests issued
four visually distinct concepts are shown together in the Choose step	PASS	4 cards (4 visible), 4 unique renders
no concept is auto-selected and Refine/Roster/Order are locked	PASS	Refine, Roster and Order disabled with aria-disabled=true
selecting a concept updates the preview, designId and unlocks downstream steps	PASS	Cosmic Energy selected, preview + designId updated, downstream unlocked
switching selection swaps cleanly with no stale state	PASS	selection switches both directions without leaking artwork or history
natural-language refinement edits the selected render	PASS	history 1, designId preserved, 5 total generation calls
colour variation recolours the current design without restarting it	PASS	palette #0A0A0A / #00A3FF / #FFFFFF, garment identity preserved
roster supports add, edit, second player and delete	PASS	Bryant #24 XL/XL x2, Carter #8 L/M x1
deleting a roster row removes only that row	PASS	deleted one row, roster now 2
order review captures customer details and validates the order	PASS	Product Brief Choose Refine Roster Order Order Review Files Team GALACTIC Qty 5 Colors Name
export panel offers PNG, SVG, PDF and ZIP for the current design	PASS	PNG, SVG, PDF, ZIP
PNG export renders the current design at the preview's aspect ratio	PASS	2400x1600 raster from the current render
Shopify checkout posts a besu:checkout payload carrying the current design	PASS	designId 1d8f6ec3-491f-4988-8b2d-624de6b1de3f, 2 items [2,3], variants 51354887192854, 51354887127318
keyboard focus reaches the concept cards and prompt input	PASS	19 focusable controls, prompt focusable
interactive controls expose labels and state	PASS	15 buttons and 5 inputs all labelled
reduced motion is honoured	PASS	shell renders with prefers-reduced-motion: reduce
Desktop (1440x900) renders the Choose step correctly	PASS	4 cards, no overflow, 6 tabs reachable
Laptop (1280x800) renders the Choose step correctly	PASS	4 cards, no overflow, 6 tabs reachable
Tablet portrait (820x1180) renders the Choose step correctly	PASS	4 cards, no overflow, 6 tabs reachable
Tablet landscape (1180x820) renders the Choose step correctly	PASS	4 cards, no overflow, 6 tabs reachable

Mobile (390x844) renders the Choose step correctly	PASS	4 cards, no overflow, 6 tabs reachable
Small mobile (360x800) renders the Choose step correctly	PASS	4 cards, no overflow, 6 tabs reachable
Mobile 375 (375x812) renders the Choose step correctly	PASS	4 cards, no overflow, 6 tabs reachable
mobile roster form is usable	PASS	3 inputs, min height 44px, no overflow
mobile order step is usable	PASS	no overflow at 390px
no uncaught page exceptions	PASS	—
no unexpected console errors	PASS	—
no failed network requests	PASS	—

21. Technical architecture

MODULE	RESPONSIBILITY
hooks/use-designer-generation.ts	Canonical generation entry points: fresh four-concept generation, refinement, colour variation, and the studio prompt router that decides between them.
lib/designer/generation-client.ts	Four art directions, the four-concept orchestration, and the single in-flight guard.
lib/designer/store.ts	Designer state, concept storage, selection semantics, version history, roster.
lib/designer/openai-service.ts	The image prompt contract for generation, refinement and colour variation.
app/api/designer/generate/route.ts	Validated generation endpoint, rate limiting, request de-duplication, image-model call and asset storage.
components/designer/concept-panel.tsx	The Choose step: the four concept cards and their selection state.
components/designer/refine-panel.tsx	Text refinement, colour presets and version history.
lib/designer/export-service.ts	PNG, SVG, production PDF and ZIP generation from the current preview.
lib/designer/shopify-service.ts	Order validation, Shopify variant mapping and the besu:checkout payload.
scripts/verify-four-concepts.mjs	Behavioural verification of the production flow.
scripts/qa-designer-flow.mjs	Real-browser QA harness, evidence capture and responsive sweep.

22. Release state

Branch	main
Verified commit	1d65304cb2981f0ab06a4591f236a5421a099f9f
Report date	August 29, 2026
Framework	Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS, HeroUI
QA browser	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) HeadlessChrome/152.0.0.0 Safari/537.36

23. Known limitations and external dependencies

1. **Live Shopify order not placed.** The `besu:checkout` payload was captured from a real browser and a real parent frame, and its contents were verified. Creating an actual cart requires the production storefront and its credentials, so the storefront listener still needs to be confirmed against this payload before go-live.
2. **The chest wordmark can still drift a letter when a design is recoloured.** Generation now spells the team name reliably, but the recolour path redraws the lettering and occasionally substitutes one character. The recommended fix — compositing the wordmark deterministically from the team-name field instead of asking the image model to draw it — is described in *Live production output* and is the clearest next improvement.
3. **Image-model output varies between briefs.** The journey was verified on two different live briefs, but generative output is not deterministic, so individual renders will always differ run to run. This is inherent to the product rather than a defect.
4. **SVG export is a raster image in an SVG wrapper.** Because the designer produces direct raster AI renders, the SVG export is not redrawable vector uniform artwork.
5. **Generation is rate limited by design.** One concept set costs four requests against a bounded per-minute, per-IP budget. Under heavy simultaneous use customers may be asked to wait a moment; this is intentional cost protection.
6. **Image-model billing must be active.** Production generation requires a valid image-model key with active credits in the deployment environment. The endpoint returns a clear customer-facing message if billing is not active.

24. Final acceptance checklist

ACCEPTANCE CRITERION	STATUS
Four independent AI uniform concepts are produced from one brief	Verified in browser
The four concepts are visibly different from each other	Verified in browser
Four distinct finished uniforms confirmed on live image-model output	Verified in browser
All four are presented in the same choice step	Verified in browser
No concept starts selected	Verified in browser
Refine, Roster and Order are locked before selection	Verified in browser
Selecting a concept makes it the main preview and working design	Verified in browser
A targeted refinement edits only the selected design	Verified in browser
Black / electric blue / white colour variation preserves the design	Verified in browser
Bryant #24 and a second player with different sizes and quantity	Verified in browser
Order review captures customer details and validates	Verified in browser
PNG, SVG, PDF and ZIP available for the current design	Verified in browser
Checkout payload retains the selected design ID and asset URL	Verified in browser
Typecheck, lint, designer test suite and production build pass	Verified
Desktop, laptop, tablet and mobile layouts free of horizontal overflow	Verified in browser
Full journey re-verified end to end on live image-model output	Verified in browser
Garment free of third-party brand marks on live output	Verified in browser
Team wordmark spelled correctly on generated concepts	Verified in browser
Team wordmark guaranteed after a recolour	Open — deterministic wordmark recommended
Live Shopify cart creation on the production storefront	Pending storefront confirmation

25. Next step

Please review the attached verified build and confirm approval. If anything in the workflow, presentation or output formats should be adjusted before sign-off, let me know and I will fold it in.

Two items remain open, and neither blocks reviewing the build:

1. Confirming that the production Shopify storefront consumes the `besu:checkout` payload documented in *Shopify order handoff*. This needs the storefront environment.
2. Making the team wordmark guaranteed rather than model-drawn, by compositing it from the team-name field as described in *Live production output*. Generation already spells it correctly; this closes the remaining drift on the recolour path. Happy to implement it on your go-ahead.

Everything else described in this report has been verified directly against the delivered build.